

RESEARCH ARTICLE

Scalar Auxiliary Variable (SAV) Stabilization of Implicit-Explicit (IMEX) Time Integration Schemes for Non-Linear Structural Dynamics

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Funding Information

US National Science Foundation, Grant/Award
Number: CNS-2229136

Abstract

Implicit-explicit (IMEX) time integration schemes are well suited for non-linear structural dynamics because of their low computational cost and high accuracy. However, the stability of IMEX schemes cannot be guaranteed for general non-linear problems. In this article, we present a scalar auxiliary variable (SAV) stabilization of high-order IMEX time integration schemes that leads to unconditional stability. The proposed IMEX-BDF k -SAV schemes treat linear terms implicitly using k th-order backward difference formulas (BDF k) and non-linear terms explicitly. This eliminates the need for iterations in non-linear problems and leads to low computational cost. Truncation error analysis of the proposed IMEX-BDF k -SAV schemes confirms that up to k th-order accuracy can be achieved and this is verified through a series of convergence tests. Unlike existing SAV schemes for first-order ordinary differential equations (ODEs), we introduce a novel SAV for the proposed schemes that allows direct solution of the second-order ODEs without transforming them into a system of first-order ODEs. Finally, we demonstrate the performance of the proposed schemes by solving several non-linear problems in structural dynamics and show that the proposed schemes can achieve high accuracy at a low computational cost while maintaining unconditional stability.

KEY WORDS

high-order time integration scheme, implicit–explicit, scalar auxiliary variable, structural dynamics, energy stability, error analysis

1 | INTRODUCTION

Direct time integration is often used to solve the second-order ordinary differential equations (ODEs) governing the dynamic response of structures and wave propagation in solids^{1–4}. Time integration is conventionally categorized into explicit and implicit schemes. Explicit schemes generally have a low computational cost per time-step and are usually used for impact, crash, and short-time wave propagation problems. However, they are almost always conditionally stable and are associated with a maximum critical time-step given by the Courant-Friedrichs-Lewy (CFL) condition⁵. On the other hand, implicit schemes permit a greater range of time-steps and several are unconditionally stable. However, this flexibility comes with a higher computational cost per time-step which is exacerbated for non-linear problems due to the need for computing the solution iteratively at each time-step. Nevertheless, implicit schemes are usually more suitable for large problems where the primary interest is in simulating the overall dynamics of a structure rather than capturing local transients. Implicit-explicit (IMEX) schemes combine implicit and explicit schemes in such a way that exploits the benefits of both types of schemes^{6,7}. IMEX schemes are well suited for non-linear problems because they do not require iterations at each time-step and have a low computational cost. However, the stability of IMEX schemes cannot be guaranteed, especially for non-linear problems⁸. In this article, we propose a stabilization scheme for IMEX schemes that guarantees unconditional stability. First, we provide a review of relevant literature on time integration schemes, especially focusing on stability for non-linear problems.

In general, time integration schemes can be classified into two groups: multi-stage methods and multi-step methods. In multi-stage methods, the time-step is divided into multiple intermediate stages and the intermediate state vectors are used to advance the solution to the next time-step. Multi-step methods, on the other hand, advance the solution by using known state vectors from several previous time-steps. Most commonly, however, one finds single-step (SS) time integration methods in the literature which lie at the intersection of multi-stage methods and multi-step methods. Examples of SS methods include the Newmark- β ⁹, Wilson- θ ¹⁰, HHT- α ¹¹, WBZ- α ¹², CH- α ¹³, CL methods¹⁴ and numerous others. Stability, accuracy and computational cost characteristics of multi-stage and multi-step methods are discussed next to provide context for the present work.

In the class of multi-stage methods, the Runge–Kutta (RK) family of integrators is widely used^{15–18}. RK methods may be explicit^{19–23} or implicit^{24–28}. In addition, multi-stage methods also include composite time integration schemes, which use different time integration schemes in different intermediate stages of each time-step^{29–43}. In general, implicit multi-stage methods can achieve high-order accuracy with unconditional stability^{44,45}. On the other hand, explicit and IMEX multi-stage methods cannot be guaranteed to be unconditionally stable for general non-linear problems^{46,47}. In addition, multi-stage methods usually have a high computational cost per time-step which, for an s -stage method, is roughly equivalent to s times the computational cost of a 1-step method.

Multi-step methods that use a linear combination of the known state vectors from previous time-steps are called linear multi-step (LMS) methods. Examples of LMS methods include Adams-Bashforth⁴⁸, Adams-Moulton⁴⁹, and backward difference formulas (BDF)⁵⁰. In LMS methods, two important results are noteworthy:

- (1) The first Dahlquist barrier^{51,52} states that the order of accuracy of a stable linear k -step method cannot exceed $k + 1$ if k is odd, or $k + 2$ if k is even.
- (2) The second Dahlquist barrier⁵³ states that there is no explicit A -stable LMS method; the highest order of accuracy of an A -stable LMS method is 2; and the trapezoidal rule (TR) has the smallest error constant among the second-order A -stable LMS methods.

From the second Dahlquist barrier, it is evident that high-order unconditionally stable LMS methods cannot be designed for linear or non-linear structural dynamics. Erlicher, Bonaventura and Bursi⁵⁴ conducted stability analysis of the generalized- α method, a second-order LMS scheme, for non-linear structural dynamics and showed that the method is energy stable in the high-frequency range.

The preceding discussion shows that ensuring the unconditional stability of high-order IMEX LMS schemes has not been achieved for general non-linear problems. However, in recent years, the scalar auxiliary variable (SAV) approach, introduced by Shen and co-workers^{55–57} for gradient flows, has been used to develop efficient high-order time integration schemes with unconditional energy stability across multiple application domains. With the SAV approach, one first defines a scalar variable Φ and augments the governing differential equations of the problem at hand with an evolution equation for this auxiliary variable which, when discretized, guarantees unconditional stability. Shen and co-workers formulated several time integrators, such as k th-order BDF (BDF k) for $1 \leq k \leq 4$ and second-order Crank-Nicolson methods, based on the SAV approach, and proved their unconditional energy stability. Note that BDF k schemes without the SAV approach are only conditionally stable when $k \geq 3$. Lin et al.⁵⁸ developed BDF k SAV schemes to solve incompressible Navier-Stokes equations. Alternative approaches using different SAVs for Navier-Stokes equations have also been formulated^{59–62}. Furthermore, Huang and Shen⁶³ developed a class of implicit-explicit (IMEX) BDF k SAV schemes for general dissipative systems, with $1 \leq k \leq 5$. They proved that the IMEX-BDF k -SAV scheme is k th-order accurate and unconditionally energy stable for dissipative systems. Other applications of the SAV approach include Schrödinger equation^{64,65}, magneto-hydrodynamics⁶⁶, electro-hydrodynamics⁶⁷, and modified phase-field crystal equation⁶⁸. However, these SAV schemes are primarily designed to solve only first-order ordinary differential equations (ODEs). While higher-order ODEs can certainly be transformed into a system of first-order ODEs and solved with the SAV approach, doing so may be computationally inefficient.

In this paper, we present a SAV stabilization of high-order IMEX BDF k time integration schemes to solve general non-linear problems in structural dynamics without converting the governing second-order ODEs to a set of first-order ODEs. Note that, there is no existing high-order IMEX time integration method in the literature that can ensure unconditional stability for general non-linear problems in structural dynamics. In Section 2, we define a scalar auxiliary variable for structural dynamics which is crucial for ensuring unconditional stability and high-order accuracy. Further, the proposed IMEX-BDF k -SAV schemes discretize linear terms using implicit time integration while treating non-linear terms explicitly, thereby avoiding the need for iterations at each time-step. We prove unconditional energy stability for linear as well as non-linear structural dynamics. We also carry out a truncation error analysis and convergence tests to show that the proposed IMEX-BDF k -SAV schemes achieve k th-order accuracy.

Finally, in Section 3, we present several numerical examples to illustrate the performance of the proposed IMEX-BDFk-SAV schemes over existing time integration methods in terms of stability, accuracy and computational cost.

2 | IMEX BDFK SAV TIME INTEGRATION SCHEMES

We begin by considering the general class of dynamical systems:

$$\frac{\partial \mathbf{x}}{\partial t} = \mathcal{D}\mathbf{x} + \mathbf{g}(\mathbf{x}) \quad (1)$$

where $\mathbf{x}$ is the unknown vector field over a spatial domain Ω , $\mathcal{D}$ is a differential operator, and $\mathbf{g}(\mathbf{x})$ is a non-linear operator. We assume that the system follows the following dissipation relationship:

$$\frac{d\Psi(\mathbf{x})}{dt} = -\Theta(\mathbf{x}) \quad (2)$$

where Ψ is an energy of the system and Θ is its dissipation. For three-dimensional structural dynamics, system (1) takes the form:

$$\begin{aligned} \dot{\mathbf{u}} &= \mathbf{v} & \text{in } \Omega \\ \rho \dot{\mathbf{v}} &= \text{div}(\boldsymbol{\sigma}) + \mathbf{b} & \text{in } \Omega \end{aligned} \quad (3)$$

where $(\dot{\cdot})$ denotes a time derivative, ρ is mass density, and $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{b}$ denote the displacement, velocity, and body force density fields, respectively, over the spatial domain of interest Ω with boundary Γ . The Cauchy stress tensor field $\boldsymbol{\sigma}$ is governed by a constitutive relation of the form $\boldsymbol{\sigma} = \boldsymbol{\sigma}(\mathbf{u}, \mathbf{v})$ which can be derived from material models and kinematics of the body²⁻⁴. Boundary conditions are given as:

$$\begin{aligned} \boldsymbol{\sigma} \mathbf{n} &= \mathbf{t}_N & \text{on } \Gamma_N \\ \mathbf{u} &= \mathbf{u}_D & \text{on } \Gamma_D \end{aligned} \quad (4)$$

where $\mathbf{t}_N$ is the applied traction on the Neumann boundary Γ_N , and $\mathbf{u}_D$ is the prescribed displacement field over the Dirichlet boundary Γ_D such that $\Gamma = \Gamma_N \cup \Gamma_D$ and $\Gamma_N \cap \Gamma_D = \emptyset$. Energy Ψ for structural dynamics is defined in terms of kinetic and potential energies as:

$$\Psi = \underbrace{\frac{1}{2} \rho \mathbf{v}^T \mathbf{v}}_{\text{Kinetic energy}} + \underbrace{W(\mathbf{u})}_{\text{Potential energy}} \quad (5)$$

where W is the energy density function that describes the constitutive material behavior.

Using the principle of virtual work and a standard spatial discretization scheme, such as finite elements²⁻⁴, Equation (3) can be converted into a system of semi-discrete second-order ordinary differential equations governing the dynamic response of a non-linear structural system:

$$\begin{aligned} \mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{f}^{\text{int}}(\mathbf{u}(t), \dot{\mathbf{u}}(t)) &= \mathbf{f}^{\text{ext}}(t) & \text{in } \Omega \quad \forall t \in [0, T] \\ \mathbf{u}(t) &= \mathbf{u}_D(t) & \text{on } \Gamma_D \quad \forall t \in (0, T] \\ \mathbf{u}(0) &= \mathbf{u}_0, \quad \dot{\mathbf{u}}(0) = \mathbf{v}_0 & \text{in } \Omega \quad \forall t = 0 \end{aligned} \quad (6)$$

where $\mathbf{M}$, $\mathbf{u}$, $\mathbf{f}^{\text{int}}$, and $\mathbf{f}^{\text{ext}}$ represent the mass matrix, nodal displacement vector, internal force vector, and external force vector, respectively. Note that the mass matrix $\mathbf{M}$ is symmetric and positive definite. The vector $\mathbf{u}_D$ is the prescribed displacement on Γ_D over the time duration $(0, T]$, and $\mathbf{u}_0$ and $\mathbf{v}_0$ denote the initial conditions for the displacement and velocity vectors, respectively.

Equation (6) governing differential equation may be solved approximately using a time integration scheme that discretizes the time of simulation $(0, T]$ into a number of time-steps Δt and advances the solution from one instant of time t_n to the next $t_{n+1} = t_n + \Delta t$ using known kinematic quantities from previous time-steps. The choice of time-step Δt is usually problem-dependent, but as a general guideline, one may pick the time-step as one-tenth of the time period corresponding to the largest frequency of interest. Note that for some time integration schemes, a special starting procedure may be needed at t_0 to compute the solution for the first few time-steps.

To facilitate the development of the proposed IMEX-BDFk-SAV schemes, the internal force vector $\mathbf{f}^{\text{int}}$ can be decomposed, without loss of generality, into a non-negative linear part and a general non-linear part, $\mathbf{f}^{\text{NL}}$, as follows:

$$\mathbf{f}^{\text{int}}(\mathbf{u}(t), \dot{\mathbf{u}}(t)) = \underbrace{\mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t)}_{\text{Non-negative linear part}} + \underbrace{\mathbf{f}^{\text{NL}}(\mathbf{u}(t), \dot{\mathbf{u}}(t))}_{\text{Non-linear part}} \quad (7)$$

where $\mathbf{C}$ and $\mathbf{K}$ matrices are defined as:

$$\mathbf{C} = \begin{cases} \mathbf{C}_0 & \text{if } \mathbf{C}_0 \succeq 0 \\ \mathbf{0} & \text{otherwise} \end{cases}, \quad \mathbf{K} = \begin{cases} \mathbf{K}_0 & \text{if } \mathbf{K}_0 \succeq 0 \\ \mathbf{0} & \text{otherwise} \end{cases} \quad (8)$$

where $\mathbf{C}_0$ and $\mathbf{K}_0$ matrices are initial tangent damping and stiffness matrices:

$$\mathbf{C}_0 = \left. \frac{\partial \mathbf{f}^{\text{int}}}{\partial \dot{\mathbf{u}}} \right|_{\mathbf{u}=\dot{\mathbf{u}}=\mathbf{0}}, \quad \mathbf{K}_0 = \left. \frac{\partial \mathbf{f}^{\text{int}}}{\partial \mathbf{u}} \right|_{\mathbf{u}=\dot{\mathbf{u}}=\mathbf{0}} \quad (9)$$

Note that matrices $\mathbf{C}$ and $\mathbf{K}$ are required to be symmetric and positive semi-definite (i.e. $\mathbf{C} \succeq 0$ and $\mathbf{K} \succeq 0$). If, for certain unstable physical systems, the initial tangent matrices $\mathbf{C}_0$ and/or $\mathbf{K}_0$ are not semi-positive definite, then they would be taken as $\mathbf{0}$.

Remark 1. Note that Equations (6) and (7) describe a variety of non-linear structural systems, e.g., non-linear elastic, elastoplastic, viscoelastic materials for small and large deformations^{2-4,69}. Even for problems with non-linear inertia terms, governing equations may be expressed using Equation (6) by linearizing the inertial term to define the mass matrix $\mathbf{M}$ and absorbing the rest into the non-linear part of the internal force $\mathbf{f}^{\text{NL}}$.

2.1 | Definition of the Scalar Auxiliary Variable (SAV) for Structural Dynamics

In structural dynamics, we first define a pseudo-energy of a general non-linear system as

$$\Psi(t) = \frac{1}{2} \dot{\mathbf{u}}(t)^T \mathbf{M} \dot{\mathbf{u}}(t) + \frac{1}{2} \mathbf{u}(t)^T \mathbf{K} \mathbf{u}(t) \quad (10)$$

Note that this pseudo-energy would be identical to the real energy for linear structural dynamics. Next, we define the scalar auxiliary variable (SAV) for general non-linear structural dynamics as:

$$\Phi(t) = \Psi(t) + \psi \quad (11)$$

where ψ is a user-defined scalar value that ensures $\Phi(t) > 0 \forall t \in [0, T]$. Note that, since pseudo-energy is always non-negative, one may choose a small non-zero value for ψ to ensure that $\Phi(t) > 0$. While such a choice is valid for maintaining unconditional stability, in general, one should choose a sufficiently large value of ψ to obtain high accuracy. As we show later in Sections 2.6 and 3, a good rule of thumb for the value for ψ is to choose it to be at least 100 times the maximum value of pseudo-energy $\Psi(t)$ expected during the entire simulation.

To formulate the IMEX-BDFk-SAV schemes, we define an evolution equation for Φ by computing its time derivative as:

$$\frac{d\Phi(t)}{dt} = \frac{d(\Psi(t) + \psi)}{dt} = -\Theta(t) \quad (12)$$

where

$$\Theta = -\dot{\mathbf{u}}^T (\mathbf{M} \ddot{\mathbf{u}} + \mathbf{K} \mathbf{u}) = \dot{\mathbf{u}}^T (\mathbf{C} \dot{\mathbf{u}} - \mathbf{f}^{\text{ext}} + \mathbf{f}^{\text{NL}}) \quad (13)$$

Note that for linear elasto-dynamics without external forces (i.e., $\mathbf{f}^{\text{NL}} = \mathbf{f}^{\text{ext}} = \mathbf{0}$), the energy in the system is dissipated (i.e., $\Theta \geq 0$) since $\mathbf{C}$ is positive semi-definite.

We now construct new IMEX-BDFk-SAV schemes by discretizing the equation of motion, Equation (6), in time using an implicit-explicit approach, where non-linear terms are treated using explicit integration (denoted with superscript EX) and linear terms are kept implicit (superscript IM):

$$\mathbf{M} \mathbf{a}_{n+1}^{\text{IM}} + \mathbf{C} \mathbf{v}_{n+1}^{\text{IM}} + \mathbf{K} \mathbf{u}_{n+1}^{\text{IM}} + \mathbf{f}^{\text{NL}}(\mathbf{u}_{n+1}^{\text{EX}}, \mathbf{v}_{n+1}^{\text{EX}}) = \mathbf{f}_{n+1}^{\text{ext}} \quad (14)$$

$$\mathbf{v}_{n+1}^{\text{IM}} = \frac{1}{\Delta t} \left\{ H^{(k)} \mathbf{u}_{n+1}^{\text{IM}} - \sum_{j=0}^{k-1} (-1)^j \frac{1}{j+1} \binom{k}{j+1} \mathbf{u}_{n-j} \right\} \quad (15)$$

$$\mathbf{a}_{n+1}^{\text{IM}} = \frac{1}{\Delta t} \left\{ H^{(k)} \mathbf{v}_{n+1}^{\text{IM}} - \sum_{j=0}^{k-1} (-1)^j \frac{1}{j+1} \binom{k}{j+1} \mathbf{v}_{n-j} \right\} \quad (16)$$

where $H^{(k)}$ is the k th harmonic number ($= \sum_{j=1}^k \frac{1}{j}$), and $\binom{k}{j+1}$ denotes the binomial coefficient ($= k! / [(k-j-1)!(j+1)!]$). Note that the explicit state vectors $\mathbf{u}_{n+1}^{EX}$ and $\mathbf{v}_{n+1}^{EX}$ are obtained using a k th-order extrapolation:

$$\mathbf{u}_{n+1}^{EX} = \sum_{j=0}^{k-1} (-1)^j \binom{k}{j+1} \mathbf{u}_{n-j}, \quad \mathbf{v}_{n+1}^{EX} = \sum_{j=0}^{k-1} (-1)^j \binom{k}{j+1} \mathbf{v}_{n-j} \quad (17)$$

Explicit state vectors $\mathbf{u}_{n+1}^{EX}$ and $\mathbf{v}_{n+1}^{EX}$ are first computed from Equation (17) and then using Equations (14)–(16), one can compute the implicit state vectors $\mathbf{u}_{n+1}^{IM}$ and $\mathbf{v}_{n+1}^{IM}$ to advance the solution. Computing these implicit state vectors is the first step of the proposed IMEX-BDFk-SAV scheme. The complete solution algorithm is summarized in Algorithm 1. Note that the implicit state vectors $\mathbf{u}_{n+1}^{IM}$ and $\mathbf{v}_{n+1}^{IM}$ would be the solution for a conventional implicit-explicit (IMEX) k th-order backward difference formulas (BDFk) scheme. However, as mentioned before, the stability of the IMEX-BDFk schemes is not guaranteed for general non-linear problems.

To ensure unconditional stability, we compute an updated value of the SAV Φ_{n+1} by discretizing its evolution equation (Equation (12)) as follows:

$$\frac{\Phi_{n+1} - \Phi_n}{\Delta t} = -\frac{\Phi_{n+1}}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \Theta[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] \quad (18)$$

where the factor $(\Phi_{n+1})/(\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi)$ on the right-hand side of the equation is obtained from the definition of the SAV in Equation (11). Note that this factor would be 1 for the time-continuous relationship in Equation (11), but in the time-discrete Equation (18), this factor ensures that $\Phi_{n+1} \geq 0$ leading to unconditional stability, as proved in Theorems 1 and 2. Next, we update the IMEX-BDFk solution using the computed value of the SAV Φ_{n+1} as:

$$\mathbf{u}_{n+1} = \Upsilon_{n+1}^{(k)} \mathbf{u}_{n+1}^{IM}, \quad \mathbf{v}_{n+1} = \Upsilon_{n+1}^{(k)} \mathbf{v}_{n+1}^{IM} \quad (19)$$

where

$$\Upsilon_{n+1}^{(k)} = 1 - (1 - \Xi_{n+1})^{\beta^{(k)}}, \quad \text{and} \quad (20)$$

$$\Xi_{n+1} = \frac{\Phi_{n+1}}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \quad (21)$$

To ensure k th-order accuracy of the scheme, parameter $\beta^{(k)}$ is chosen as:

$$\beta^{(k)} = \begin{cases} 1 + (k+1)/2 & \text{for odd numbers of } k \\ 1 + k/2 & \text{for even numbers of } k \end{cases} \quad (22)$$

Truncation error analysis (see Section 2.5 for details) confirms k th-order accuracy of the proposed IMEX-BDFk-SAV schemes. Note that the displacement and velocity state vectors ($\mathbf{u}_{n+1}$ and $\mathbf{v}_{n+1}$) are updated using the same scalar coefficient $\Upsilon_{n+1}^{(k)}$ which helps in keeping the computational overhead of the SAV approach low and also leads to an unconditionally stable scheme (see Section 2.4 for details). Finally, if needed, the acceleration $\mathbf{a}_{n+1}$ may be updated from the governing equation of motion.

Remark 2. In the formulation above, matrices $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ do not change with time. Thus, one needs to factorize the matrix operator only once at the beginning of the simulation. Further, note that the non-linear part of the internal force, $\mathbf{f}^{NL}$, is computed using explicit *predicted* state vectors $\mathbf{u}_{n+1}^{EX}$ and $\mathbf{v}_{n+1}^{EX}$ and solution of the implicit *corrected* state vectors $\mathbf{u}_{n+1}^{IM}$ and $\mathbf{v}_{n+1}^{IM}$ only involves the linear part of the internal force. Thus, there is no need for non-linear iterations such as those in conventional implicit time-stepping methods. Similarly, there are no iterations needed for the SAV stabilization either, because Ψ and Θ in Equation (18) use implicit state vectors that are already computed in Equations (14)–(16).

2.2 | Solution Recovery From Instability After Temporal Locking

Even though the proposed IMEX-BDFk-SAV method, as presented above, is unconditionally stable, it suffers from a drawback that when the solution of underlying IMEX-BDFk scheme diverges due to instability (i.e., $\|\mathbf{u}_{n+1}^{IM}\| \rightarrow \infty$ and $\|\mathbf{v}_{n+1}^{IM}\| \rightarrow \infty$), the computed solution from the IMEX-BDFk-SAV scheme tends to zero (i.e., $\|\mathbf{u}_{n+1}\| \rightarrow 0$ and $\|\mathbf{v}_{n+1}\| \rightarrow 0$). This occurs because,

when $\|\mathbf{u}_{n+1}^{IM}\| \rightarrow \infty$ and $\|\mathbf{v}_{n+1}^{IM}\| \rightarrow \infty$, the definition of Ψ in Equation (10) leads to

$$\frac{\|\mathbf{u}_{n+1}^{IM}\|}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}]} \rightarrow 0, \quad \frac{\|\mathbf{v}_{n+1}^{IM}\|}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}]} \rightarrow 0 \quad (23)$$

This leads to $\Xi_{n+1} \rightarrow 0$ and $\Upsilon_{n+1}^{(k)} \rightarrow 0$ and consequently, $\|\mathbf{u}_{n+1}\|, \|\mathbf{v}_{n+1}\| \rightarrow 0$.

The fact that $\|\mathbf{u}_{n+1}\|, \|\mathbf{v}_{n+1}\| \rightarrow 0$ by itself is not an issue, but for some problems, when Φ_{n+1} is also small, then the computed solution for the following time-steps continues to remain close to zero (i.e., $\|\mathbf{u}_{n+m}\|, \|\mathbf{v}_{n+m}\| \rightarrow 0$ for any value of m). We call this phenomenon *temporal locking* where the structure becomes almost rigid and gives an almost zero response even under the action of external forces. This temporal locking phenomenon is demonstrated in the numerical example presented in Section 3.4.

To understand why temporal locking occurs, we consider the solution at the next time-step, t_{n+2} , after $\mathbf{u}_{n+1}, \mathbf{v}_{n+1}, \Phi_{n+1}$ have all been computed. Following Equations (14)–(18), $\mathbf{u}_{n+2}^{IM}, \mathbf{v}_{n+2}^{IM}, \Phi_{n+2}$ are obtained. Next, using Equation (21), we can show that:

$$\Xi_{n+2} = \frac{\Phi_{n+2}}{\Psi[\mathbf{u}_{n+2}^{IM}, \mathbf{v}_{n+2}^{IM}] + \psi} \leq \frac{\Phi_{n+2}}{\psi} \leq \frac{2\Phi_{n+1}}{\psi} \approx 0 \quad (24)$$

In the above, we make use of the Gronwall Lemma which states that $\Phi_{n+2} \leq 2\Phi_{n+1}$ for general non-linear systems (see Equation (44)) and since $\Phi_{n+1} \approx 0$, thus $\Xi_{n+2} \rightarrow 0$ and $\Upsilon_{n+2}^{(k)} \rightarrow 0$. Thus, from Equations (19), (20), and (24), $\|\mathbf{u}_{n+2}\| \approx 0$ and $\|\mathbf{v}_{n+2}\| \approx 0$. This phenomenon continues over the following time-steps as well making $\|\mathbf{u}_{n+m}\| \approx 0$ and $\|\mathbf{v}_{n+m}\| \approx 0$ for any value of m .

To avoid the temporal locking, for the proposed IMEX-BDF k -SAV schemes, we set a minimum value of Φ_{n+1} as $\varepsilon_{\text{tol}}\psi$, where ε_{tol} is a small number (in this study, ε_{tol} is chosen as 10^{-7}). Note that the minimum value of Φ_{n+1} is problem-dependent and that is why it is scaled with the variable ψ . Thus, when $\|\mathbf{u}_{n+1}\| \approx 0, \|\mathbf{v}_{n+1}\| \approx 0$ and $\Phi_{n+1}/\psi < \varepsilon_{\text{tol}}$, then we initiate a recovery of the SAV, Φ , by replacing the value of Φ_{n+1} with the following expression:

$$\Phi_{n+1} = \Psi[\mathbf{u}_{n+1}, \mathbf{v}_{n+1}] + \psi = \left(\Upsilon_{n+1}^{(k)}\right)^2 \Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi \quad (25)$$

which is based on the definition of the SAV in Equation (11) rather than its updated value computed from Equation (18). This ensures that the energy of the system, Ψ_{n+1} , and consequently the SAV, Φ_{n+1} , maintain their values as the system evolves. Note that, since $\Upsilon_{n+1}^{(k)}$ and $\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}]$ is already computed at each time-step, the computational cost for this recovery is negligible. Further, since the updated SAV Φ_{n+1} is always bounded (because $\mathbf{u}_{n+1}$ and $\mathbf{v}_{n+1}$ are bounded), the stability analysis presented in Section 2.4 remains valid. Recovering the value of Φ_{n+1} , as shown above, simply amounts to a modification of the state of the system for the following time-steps. One can imagine this recovery as dividing the time interval of simulation into two segments, one before and one after the recovery. The part of the simulation after the recovery can then be simply thought of as a new simulation with an updated initial condition. Lastly, the local truncation error (LTE) analysis presented in Section 2.5 also remains valid because we conduct the LTE analysis only going from t_n to t_{n+1} .

2.3 | Starting Procedure and Algorithm

A special starting procedure is needed for the proposed IMEX-BDF k -SAV schemes with $k \geq 2$ to achieve k th-order accuracy. At the initial time t_0 , the state vectors for the first $(k-1)$ steps must be computed with a method that is at least $(k-1)$ th-order accurate. In this study, we use the $(k-1)$ th-order Runge–Kutta method for the first $(k-1)$ steps.

We summarize the procedure of the proposed IMEX-BDF k -SAV schemes for structural dynamics in Algorithm 1.

2.4 | Stability Analysis

In this section, we prove that the proposed IMEX-BDF k -SAV schemes are unconditionally energy stable, first for linear structural dynamics in Theorem 1 and subsequently for non-linear structural dynamics in Theorem 2.

Theorem 1. *For linear structural dynamics, in the absence of external forces $\mathbf{f}^{\text{ext}}$, the IMEX-BDF k -SAV schemes for any $k \geq 1$ are unconditionally energy stable such that*

$$\Phi_{n+1} \leq \Phi_n \quad (26)$$

Algorithm 1 Time stepping for the IMEX-BDF k -SAV schemes.

1. Initial calculation
 - a) Construct the mass, stiffness, and damping matrices $\mathbf{M}$, $\mathbf{K}$, and $\mathbf{C}$
 - b) Initialize displacement and velocity vectors, and SAV: $\mathbf{u}_0 = \mathbf{u}_0^{EX} = \mathbf{u}(0)$, $\mathbf{v}_0 = \mathbf{v}_0^{EX} = \dot{\mathbf{u}}(0)$, $\Phi_0 = \Psi[\mathbf{u}_0, \mathbf{v}_0] + \psi$
2. Starting procedure only when $k \geq 2$ ($0 \leq n < k-1$)
 - a) Compute $\mathbf{u}_{n+1}$ and $\mathbf{v}_{n+1}$ using $(k-1)^{\text{th}}$ -order time integration schemes (e.g. Runge-Kutta method)
 - b) If $n = k-2$, then compute the SAV: $\Phi_{n+1} = \Psi[\mathbf{u}_{n+1}, \mathbf{v}_{n+1}] + \psi$
3. For each cycle of the time-step ($n \geq k-1$)
 - a) Compute the explicit state vectors $\mathbf{u}_{n+1}^{EM}$ and $\mathbf{v}_{n+1}^{EM}$ from Equation (17).
 - b) Compute the implicit state vectors $\mathbf{u}_{n+1}^{IM}$ and $\mathbf{v}_{n+1}^{IM}$ from Equations (14)-(16).
 - c) Compute the SAV Φ_{n+1} from Equation (18).
 - d) Update the state vectors $\mathbf{u}_{n+1}$ and $\mathbf{v}_{n+1}$ from Equations (19)-(21).
 - e) If $\Phi_{n+1}/\psi < \varepsilon_{\text{tol}}$, then recover the SAV Φ_{n+1} from Equation (25)

Furthermore, there exists $U^{(k)} > 0$ such that

$$\Psi[\mathbf{u}_n, \mathbf{v}_n] \leq (U^{(k)})^2 \quad \forall n \in [0, T/\Delta t] \quad (27)$$

Proof. For linear structural dynamics in the absence of external forces, the definition of Θ in Equation (13) ensures that $\Theta[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] \geq 0$ at all times. Further, from the definition of the SAV Φ in Equation (11), $\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi > 0$. Thus, given $\Phi_n \geq 0$, Equation (18) may be rewritten as:

$$\Phi_{n+1} = \frac{\Phi_n}{1 + \Delta t \frac{\Theta[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}]}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi}} \quad (28)$$

ensuring that $\Phi_{n+1} \geq 0$. Next, multiplying Equation (18) by Δt and noting that the factor on the right-hand side of Equation (18) is simply Ξ_{n+1} , as defined in Equation (21), we conclude that:

$$\Phi_{n+1} - \Phi_n = -\Delta t \Xi_{n+1} \Theta[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] \leq 0 \quad (29)$$

because, from Equation (21), $\Xi_{n+1} \geq 0$. This proves Equation (26) of this theorem.

To prove Equation (27), let $U := \Phi_0$. Then, Equation (26) ensures that $\Phi_{n+1} \leq U$. From Equation (21),

$$\Xi_{n+1} = \frac{\Phi_{n+1}}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \leq \frac{U}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \quad (30)$$

From Equation (20), the scalar coefficient $\Upsilon_{n+1}^{(k)}$ can be represented as $\Upsilon_{n+1}^{(k)} = \Xi_{n+1} P_{\beta^{(k)}-1}(\Xi_{n+1})$, where $P_{\beta^{(k)}-1}$ is a polynomial of degree $\beta^{(k)} - 1$. Note that the absolute value of the polynomial $P_{\beta^{(k)}-1}$ is bounded. Further, since $U > 0$ and $\psi > 0$, there exists $U^{(k)} > 0$ such that

$$|P_{\beta^{(k)}-1}(\Xi_{n+1})| \leq \frac{2\sqrt{\psi}}{U} U^{(k)} \quad (31)$$

From Equations (30) and (31),

$$|\Upsilon_{n+1}^{(k)}| = |\Xi_{n+1} P_{\beta^{(k)}-1}(\Xi_{n+1})| \leq \frac{2\sqrt{\psi} U^{(k)}}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \quad (32)$$

From Equations (10) and (19), the pseudo-energy Ψ at t_{n+1} may be written as:

$$\Psi[\mathbf{u}_{n+1}, \mathbf{v}_{n+1}] = \left(\Upsilon_{n+1}^{(k)} \right)^2 \Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] \quad (33)$$

Substituting Equation (32) into Equation (33), the following inequality is obtained:

$$\Psi[\mathbf{u}_{n+1}, \mathbf{v}_{n+1}] \leq \left(\frac{2\sqrt{\psi} U^{(k)}}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \right)^2 \Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] \leq (U^{(k)})^2 \quad (34)$$

Note that we use the following identity in the last step above:

$$\frac{4\psi\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}]}{(\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi)^2} \leq 1 \quad (35)$$

which in turn is obtained from the following statement: $(\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] - \psi)^2 = (\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi)^2 - 4\psi\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] \geq 0$. The proof is complete. $\square$

Theorem 2. For non-linear structural dynamics with bounded non-linear force $\mathbf{f}^{NL}$ and in the absence of external forces $\mathbf{f}^{ext}$, given $\psi \geq 2\Delta t^2 \|\mathbf{L}^{-1}\mathbf{f}^{NL}\|^2$ for all t , where $\mathbf{M} = \mathbf{L}\mathbf{L}^T$ (Cholesky decomposition), there exists $U^{(k)} > 0$ such that

$$\Psi[\mathbf{u}_n, \mathbf{v}_n] \leq (U^{(k)})^2 \quad \forall n \in [0, T/\Delta t] \quad (36)$$

Proof. For non-linear structural dynamics in the absence of external forces, Equation (13) states:

$$\Theta[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] = \mathbf{v}_{n+1}^{IM^T} (\mathbf{C}\mathbf{v}_{n+1}^{IM} + \mathbf{f}^{NL}(\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM})) \quad (37)$$

Thus,

$$|\Theta[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}]| \leq |\mathbf{v}_{n+1}^{IM^T} \mathbf{f}^{NL}(\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM})| \leq |\mathbf{v}_{n+1}^{IM^T} \mathbf{L}\mathbf{L}^{-1} \mathbf{f}^{NL}(\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM})| \leq \|\mathbf{L}^T \mathbf{v}_{n+1}^{IM}\| \|\mathbf{L}^{-1} \mathbf{f}^{NL}(\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM})\| \quad (38)$$

From Equation (10) and choosing $\psi \geq 2\Delta t^2 \|\mathbf{L}^{-1} \mathbf{f}^{NL}\|^2$,

$$\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi \geq \frac{1}{2} \mathbf{v}_{n+1}^{IM^T} \mathbf{M} \mathbf{v}_{n+1}^{IM} + \psi \geq \frac{1}{2} \mathbf{v}_{n+1}^{IM^T} \mathbf{M} \mathbf{v}_{n+1}^{IM} + 2\Delta t^2 \|\mathbf{L}^{-1} \mathbf{f}^{NL}\|^2 \geq \frac{1}{2} \mathbf{v}_{n+1}^{IM^T} \mathbf{L}\mathbf{L}^T \mathbf{v}_{n+1}^{IM} + 2\Delta t^2 \|\mathbf{L}^{-1} \mathbf{f}^{NL}\|^2 \quad (39)$$

Dividing inequality (38) by inequality (39), we obtain:

$$\left| \frac{\Theta[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}]}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \right| \leq \frac{\|\mathbf{L}^T \mathbf{v}_{n+1}^{IM}\| \|\mathbf{L}^{-1} \mathbf{f}^{NL}(\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM})\|}{\frac{1}{2} \|\mathbf{L}^T \mathbf{v}_{n+1}^{IM}\|^2 + 2\Delta t^2 \|\mathbf{L}^{-1} \mathbf{f}^{NL}(\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM})\|^2} \leq \frac{1}{2\Delta t} \quad (40)$$

where, in the last step above, we use the fact:

$$(\|\mathbf{L}^T \mathbf{v}_{n+1}^{IM}\| - 2\Delta t \|\mathbf{L}^{-1} \mathbf{f}^{NL}(\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM})\|)^2 \geq 0 \quad (41)$$

Thus, given $\Phi_n > 0$, Equation (28) and inequality (40) lead to:

$$\Phi_{n+1} = \Phi_n \left(1 + \Delta t \frac{\Theta[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}]}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \right)^{-1} > 0 \quad (42)$$

Then, summing up Equation (18) from $n = 0$ to m ,

$$\Phi_{m+1} = \Phi_0 - \Delta t \sum_{j=0}^m \frac{\Phi_{j+1}}{\Psi[\mathbf{u}_{j+1}^{IM}, \mathbf{v}_{j+1}^{IM}] + \psi} \Theta[\mathbf{u}_{j+1}^{IM}, \mathbf{v}_{j+1}^{IM}] \leq \Phi_0 + \Delta t \sum_{j=0}^m \left| \frac{\Theta[\mathbf{u}_{j+1}^{IM}, \mathbf{v}_{j+1}^{IM}]}{\Psi[\mathbf{u}_{j+1}^{IM}, \mathbf{v}_{j+1}^{IM}] + \psi} \right| \Phi_{j+1} \leq \Phi_0 + \frac{1}{2} \sum_{j=0}^m \Phi_{j+1} \quad (43)$$

Applying the discrete Gronwall lemma (see Clark⁷⁰), it can be proved that

$$\Phi_m \leq 2^m \Phi_0 \quad \forall m \in [0, T/\Delta t] \quad (44)$$

which implies that

$$\Xi_{n+1} = \frac{\Phi_{n+1}}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \leq \frac{2^{n+1} \Phi_0}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \quad (45)$$

Recall $\Upsilon_{n+1}^{(k)} = \Xi_{n+1} P_{\beta^{(k)}-1}(\Xi_{n+1})$. Since $\Psi_0 > 0$ and $\psi > 0$, there exists $U^{(k)} > 0$ such that

$$|P_{\beta^{(k)}-1}(\Xi_{n+1})| \leq \frac{2\sqrt{\psi}}{2^{n+1} \Phi_0} U^{(k)} \quad (46)$$

From Equations (45) and (46),

$$|\Upsilon_{n+1}^{(k)}| = |\Xi_{n+1} P_{\beta^{(k)}-1}(\Xi_{n+1})| \leq \frac{2\sqrt{\psi} U^{(k)}}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \quad (47)$$

From Equation (19), Equation (47) implies

$$\Psi[\mathbf{u}_{n+1}, \mathbf{v}_{n+1}] = \left(\Upsilon_{n+1}^{(k)} \right)^2 \Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] \leq \left(\frac{2\sqrt{\psi}U^{(k)}}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi} \right)^2 \Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] \leq (U^{(k)})^2 \quad (48)$$

where, in the final step above, Equation (35) is used. The proof is complete. $\square$

2.5 | Local Truncation Error Analysis

In this section, we analyze the local truncation error (LTE) of the proposed IMEX-BDF k -SAV schemes in linear structural dynamics. Note that for linear structural dynamics, the governing second-order ODEs are obtained from Equation (6) where:

$$\mathbf{f}^{\text{int}} = \mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) \quad (49)$$

In this case, the definition of Θ in Equation (13) simplifies to:

$$\Theta = \dot{\mathbf{u}}^T (\mathbf{C}\dot{\mathbf{u}} - \mathbf{f}^{\text{ext}}) \quad (50)$$

As the solutions for displacement and velocity vectors, $\mathbf{u}_{n+1}$ and $\mathbf{v}_{n+1}$ respectively, are computed by advancing from t_n to t_{n+1} , the LTEs in these quantities can be obtained as:

$$\tau_{\mathbf{u}} = \mathbf{u}_{n+1} - \mathbf{u}(t_{n+1}), \quad \tau_{\mathbf{v}} = \mathbf{v}_{n+1} - \dot{\mathbf{u}}(t_{n+1}) \quad (51)$$

where $\mathbf{u}(t_{n+1})$ and $\dot{\mathbf{u}}(t_{n+1})$ denote the exact displacement and velocity vectors. The exact solution can be obtained from a Taylor series expansion about t_n as follows:

$$\mathbf{u}(t_{n+1}) = \mathbf{u}(t_n) + \sum_{k=1}^{\infty} \frac{\Delta t^k}{k!} \frac{d^k \mathbf{u}(t_n)}{dt^k}, \quad \dot{\mathbf{u}}(t_{n+1}) = \dot{\mathbf{u}}(t_n) + \sum_{k=1}^{\infty} \frac{\Delta t^k}{k!} \frac{d^k \dot{\mathbf{u}}(t_n)}{dt^k} \quad (52)$$

Note that, for LTE, we assume that $\mathbf{u}(t_n) = \mathbf{u}_n$ and $\dot{\mathbf{u}}(t_n) = \mathbf{v}_n$ are exact displacement and velocity vectors at t_n . In this section, the external force vector is assumed to be continuous in time, then $\mathbf{f}^{\text{ext}}(t_{n+1})$ can be expressed as follows:

$$\mathbf{f}^{\text{ext}}(t_{n+1}) = \mathbf{f}^{\text{ext}}(t_n) + \sum_{k=1}^{\infty} \frac{\Delta t^k}{k!} \frac{d^k \mathbf{f}^{\text{ext}}(t_n)}{dt^k} \quad (53)$$

Using Equations (52) and (53), it can be shown that the conventional BDF k schemes are k th-order accurate (see Iserles⁴⁴ for details):

$$\mathbf{u}_{n+1}^{IM} - \mathbf{u}(t_{n+1}) \propto C_u \Delta t^{k+1} + O(\Delta t^{k+2}), \quad \mathbf{v}_{n+1}^{IM} - \dot{\mathbf{u}}(t_{n+1}) \propto C_v \Delta t^{k+1} + O(\Delta t^{k+2}) \quad (54)$$

where C_u and C_v are constant coefficients of the leading-order terms in the LTEs in displacement and velocity vectors, respectively. To compute the LTEs for the proposed IMEX-BDF k -SAV schemes, Ξ_{n+1} is obtained by substituting Equation (28) into Equation (21) and using the definition of Θ in Equation (50):

$$\Xi_{n+1} = \frac{\Phi_n}{\Psi[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}] + \psi + \Delta t \Theta[\mathbf{u}_{n+1}^{IM}, \mathbf{v}_{n+1}^{IM}]} = \frac{\mathbf{v}_n^T \mathbf{M} \mathbf{v}_n + \mathbf{u}_n^T \mathbf{K} \mathbf{u}_n + 2\psi}{\mathbf{v}_{n+1}^{IM^T} \mathbf{M} \mathbf{v}_{n+1}^{IM} + \mathbf{u}_{n+1}^{IM^T} \mathbf{K} \mathbf{u}_{n+1}^{IM} + 2\psi + 2\Delta t \mathbf{v}_{n+1}^{IM^T} (\mathbf{C} \mathbf{v}_{n+1}^{IM} - \mathbf{f}^{\text{ext}}(t_{n+1}))} \quad (55)$$

Using Equation (54), Equation (55) is rewritten as:

$$\Xi_{n+1} \propto \frac{\mathbf{v}_n^T \mathbf{M} \mathbf{v}_n + \mathbf{u}_n^T \mathbf{K} \mathbf{u}_n + 2\psi}{\dot{\mathbf{u}}(t_{n+1})^T \mathbf{M} \dot{\mathbf{u}}(t_{n+1}) + \mathbf{u}(t_{n+1})^T \mathbf{K} \mathbf{u}(t_{n+1}) + 2\psi + 2\Delta t \dot{\mathbf{u}}(t_{n+1})^T (\mathbf{C} \dot{\mathbf{u}}(t_{n+1}) - \mathbf{f}^{\text{ext}}(t_{n+1})) + O(\Delta t^{k+1})} \quad (56)$$

Using Equations (52) and (53), the following terms in denominator of Equation (56) can be written as:

$$\begin{aligned} & \dot{\mathbf{u}}(t_{n+1})^T \mathbf{M} \dot{\mathbf{u}}(t_{n+1}) + \mathbf{u}(t_{n+1})^T \mathbf{K} \mathbf{u}(t_{n+1}) + 2\Delta t \dot{\mathbf{u}}(t_{n+1})^T (\mathbf{C} \dot{\mathbf{u}}(t_{n+1}) - \mathbf{f}^{\text{ext}}(t_{n+1})) \\ & \propto \mathbf{v}_n^T \mathbf{M} \mathbf{v}_n + \mathbf{u}_n^T \mathbf{K} \mathbf{u}_n + 2\Delta t \mathbf{v}_n^T (\mathbf{M} \mathbf{a}_n + \mathbf{C} \mathbf{v}_n + \mathbf{K} \mathbf{u}_n - \mathbf{f}^{\text{ext}}(t_n)) + C_E \Delta t^2 + O(\Delta t^3) \\ & \propto \mathbf{v}_n^T \mathbf{M} \mathbf{v}_n + \mathbf{u}_n^T \mathbf{K} \mathbf{u}_n + C_E \Delta t^2 + O(\Delta t^3) \end{aligned} \quad (57)$$

where C_E is the coefficient of the leading-order term and $\mathbf{M}\mathbf{a}_n + \mathbf{C}\mathbf{v}_n + \mathbf{K}\mathbf{u}_n = \mathbf{f}^{\text{ext}}(t_n)$. Thus, Equation (56) can be simplified to:

$$\Xi_{n+1} \propto \frac{\mathbf{v}_n^T \mathbf{M} \mathbf{v}_n + \mathbf{u}_n^T \mathbf{K} \mathbf{u}_n + 2\psi}{\mathbf{v}_n^T \mathbf{M} \mathbf{v}_n + \mathbf{u}_n^T \mathbf{K} \mathbf{u}_n + 2\psi + C_E \Delta t^2 + O(\Delta t^3)} \propto 1 + \frac{C_E}{2\Phi_n} \Delta t^2 + O(\Delta t^3) \quad (58)$$

where $\Phi_n = \frac{1}{2} (\mathbf{v}_n^T \mathbf{M} \mathbf{v}_n + \mathbf{u}_n^T \mathbf{K} \mathbf{u}_n) + \psi$. Since $\Upsilon_{n+1}^{(k)} = 1 - (1 - \Xi_{n+1})^{\beta^{(k)}}$, Equation (58) implies:

$$\Upsilon_{n+1}^{(k)} \propto 1 - \left(-\frac{C_E}{2\Phi_0} \right)^{\beta^{(k)}} \Delta t^{2\beta^{(k)}} \quad (59)$$

Finally, from Equations (19), (54), and (59), LTEs in displacement and velocity vectors are obtained as:

$$\tau_{\mathbf{u}} = \mathbf{u}_{n+1} - \mathbf{u}(t_{n+1}) = \Upsilon_{n+1}^{(k)} \mathbf{u}_{n+1}^{IM} - \mathbf{u}(t_{n+1}) \propto C_u \Delta t^{k+1} - \left(-\frac{C_E}{2\Phi_0} \right)^{\beta^{(k)}} \mathbf{u}(t_{n+1}) \Delta t^{2\beta^{(k)}} \quad (60)$$

$$\tau_{\mathbf{v}} = \mathbf{v}_{n+1} - \dot{\mathbf{u}}(t_{n+1}) = \Upsilon_{n+1}^{(k)} \mathbf{v}_{n+1}^{IM} - \dot{\mathbf{u}}(t_{n+1}) \propto C_v \Delta t^{k+1} - \left(-\frac{C_E}{2\Phi_0} \right)^{\beta^{(k)}} \dot{\mathbf{u}}(t_{n+1}) \Delta t^{2\beta^{(k)}} \quad (61)$$

Since $2\beta^{(k)} > k + 1$, the proposed IMEX-BDF k -SAV schemes leads to k th-order accuracy. Thus, the LTEs for the underlying BDF k schemes are preserved:

$$\tau_{\mathbf{u}} \propto C_u \Delta t^{k+1} + O(\Delta t^{k+2}), \quad \tau_{\mathbf{v}} \propto C_v \Delta t^{k+1} + O(\Delta t^{k+2}) \quad (62)$$

It is important to note that with an increase in $\beta^{(k)}$, the energy bound $(U^{(k)})^2$ also increases (see Equations (31) and (46)). Therefore, as given in Equation (22), we recommend selecting the smallest $\beta^{(k)}$ such that $2\beta^{(k)} > k + 1$. This is in contrast to the value $\beta^{(k)} = k + 1$ suggested by Huang and Shen⁶³ for general dissipative systems because, in their derivation, Ξ_{n+1} is a first-order approximation to 1. As shown in Equation (58), Ξ_{n+1} for the proposed IMEX-BDF k -SAV schemes is a second-order approximation to 1 for any value of k . Thus, for structural dynamics, we can use a value of $\beta^{(k)}$ that is about half of the value used for general dissipative systems. This leads to a lower energy bound $(U^{(k)})^2$ and consequently better performance of the proposed IMEX-BDF k -SAV schemes.

To verify the theoretical LTEs found above, we compute the LTEs for a typical modal equation of motion with linear internal forces:

$$\ddot{u}(t) + 2\zeta\omega_0\dot{u}(t) + \omega_0^2 u(t) = f^{\text{ext}}(t) \quad (63)$$

$$u(0) = u_0, \quad \dot{u}(0) = v_0 \quad (64)$$

where ω_0 , ζ , u , and f^{ext} are the natural frequency, damping ratio, modal displacement, and modal external force, respectively. We consider a sinusoidal external forcing function: $f^{\text{ext}}(t) = p_0 \sin \omega_f t$, where p_0 and ω_f are the amplitude and excitation frequency, respectively. Note that the exact solution to this problem is given as:

$$u(t) = e^{-\zeta\omega_0 t} (A_f \cos(\omega_d t) + B_f \sin(\omega_d t)) + \frac{p_0}{\eta} ((\omega_0^2 - \omega_f^2) \sin(\omega_f t) - 2\omega_0\omega_f\zeta \cos(\omega_f t)) \quad (65)$$

where $A_f = u_0 + 2\frac{p_0}{\eta}\zeta\omega_0\omega_f$, $B_f = \frac{v_0 + \zeta\omega_0 u_0}{\omega_d} + \frac{p_0}{\eta} \frac{\omega_f}{\omega_d} (\omega_f^2 - \omega_0^2 + 2\zeta^2\omega_0^2)$, and $\eta = (\omega_0^2 - \omega_f^2)^2 + (2\zeta\omega_0\omega_f)^2$. Using the exact solution above and the analysis in Section 2.5, the LTEs in displacement and velocity for the modal equation of motion are computed and reported in Appendix A. Note that the LTEs in displacements and velocities are found to be of the order of Δt^{k+1} for $1 \leq k \leq 5$. Thus, the global errors in displacements and velocities can be expected to be of the order of Δt^k for $1 \leq k \leq 5$.

2.6 | Convergence Tests

To verify the theoretical findings in Section 2.5, we compute global errors for the proposed IMEX-BDF k -SAV schemes when solving linear and non-linear problems. To quantify global errors, we define the following error norm:

$$\mathcal{E}(\mathbf{z}) = \sqrt{\sum_{n=0}^{T/\Delta t} (z_n - \tilde{z}_n)^2} / \sqrt{\sum_{n=0}^{T/\Delta t} (\tilde{z}_n)^2} \quad (66)$$

where $\mathbf{z}$ may be displacement, velocity or acceleration and $\tilde{\mathbf{z}}$ denotes the exact solution or a reference solution, when an exact solution is not available.

2.6.1 | Linear Structural Dynamics

We consider a damped single-degree-of-freedom (SDOF) problem, as given in Equation (63), with damping ratio $\zeta = 0.2$, natural frequency $\omega_0 = 2\pi$, and the corresponding natural period $T_0 = 1$. The duration of the simulation is taken to be $[0, T]$, where $T = 10$.

We apply zero initial displacement and velocity and measure the global error in the forced response under a load given as $f^{\text{ext}}(t) = \sin 2\omega_0 t$. For the proposed IMEX-BDF k -SAV schemes, we need an estimate of the maximum energy Ψ for this problem, which can be obtained by assuming that the maximum displacement will be twice the static response i.e. $u_{\max} \approx 2p_0/\omega_0^2$. Thus, the maximum energy stored in the underlying undamped system would be $\Psi_{\max} \approx \frac{1}{2}\omega_0^2 u_{\max}^2 \approx 0.05$. Following the guideline for the choice of SAV in Equation (11), we choose $\psi = 100\Psi_{\max} = 5$. Figure 1 shows the global error in all kinematic quantities using the proposed IMEX-BDF k -SAV schemes with $\psi = 5$. For comparison, the global errors from the standard Bathe method with $\gamma = 0.5$ ^{29,30}, Newmark trapezoidal rule (TR)⁹, and the generalized- α method with $\rho_\infty = 0$ ¹³ are also shown. From these plots, we can verify that the proposed IMEX-BDF k -SAV schemes are k th-order accurate in displacement, velocity, and acceleration for linear problems in structural dynamics.

To investigate the effect of ψ , we plot the global errors for different values of $\psi/\Psi_{\max}$ for two values of $\Delta t/T_0$ in Figure 2. We observe that global errors do depend on the value of ψ , however, as for $\psi/\Psi_{\max}$ greater than 100, the global errors stabilize.

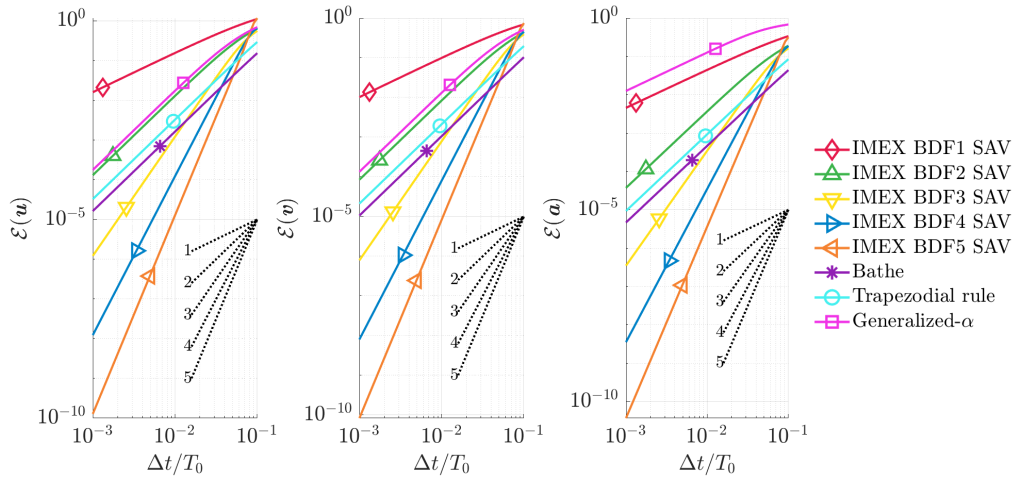


FIGURE 1 Error norms for the damped SDOF problem with $f^{\text{ext}} = \sin 2\omega_0 t$ and $u_0 = v_0 = 0$.

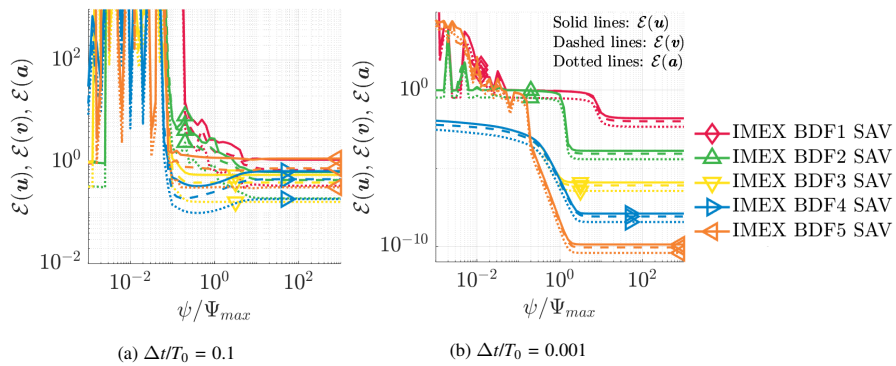


FIGURE 2 Error norms for the damped SDOF problem with $f^{\text{ext}} = \sin 2\omega_0 t$ and $u_0 = v_0 = 0$ using the proposed BDF k SAV schemes for various values of $\psi/\Psi_{\max} \in [10^{-3}, 10^3]$ at intervals of $10^{0.1}$.

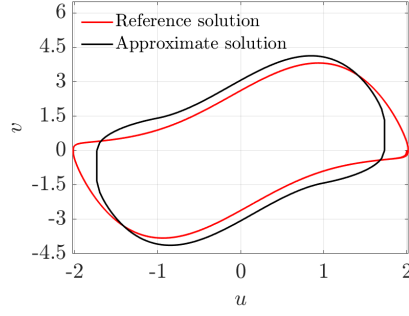


FIGURE 3 Phase portrait of the SDOF Van der Pol problem with $f^{\text{ext}}(t) = 0$, $u_0 = 2$, and $v_0 = 0$.

2.6.2 | Non-linear Structural Dynamics

Here, we consider two types of non-linear SDOF problems: a Van der Pol oscillator and a Duffing oscillator. The governing equations of these problems are expressed as

$$\text{Van der Pol oscillator: } m\ddot{u} + k_1 u - \mu(1 - u^2)\dot{u} = f^{\text{ext}} \quad (67)$$

$$\text{Duffing oscillator: } m\ddot{u} + c\dot{u} + k_1 u + k_3 u^3 = f^{\text{ext}} \quad (68)$$

First, we consider the Van der Pol oscillator with $m = 1$, $k_1 = 1$, $\mu = 2$, and $f^{\text{ext}} = 0$ over the time duration $[0, T]$, where $T = 15$. A non-zero initial displacement and a zero initial velocity are applied: $u_0 = 2$ and $v_0 = 0$. As estimate of the maximum pseudo-energy Ψ is obtained from an approximate solution $\ddot{u}$ and $\ddot{v}$ of a limit cycle for any value of $\mu > 0$ (see Lopez and Lopez-Ruiz⁷¹ for details):

$$\ddot{v}(\ddot{u}) = \begin{cases} \sqrt{3 - \ddot{u}^2} & \text{if } -\sqrt{3} < \ddot{u} < -1 \\ \sqrt{3 - \ddot{u}^2} + \mu \left(-\frac{1}{3}\ddot{u}^3 + \ddot{u} + \frac{2}{3} \right) & \text{if } -1 < \ddot{u} < \sqrt{3} \end{cases} \quad (69)$$

This approximate solution is compared to a reference solution for the Van der Pol oscillator in Figure 3. Using Equation (69), the pseudo-energy Ψ is approximated as $\frac{1}{2}m\ddot{v}^2 + \frac{1}{2}k_1\ddot{u}^2$ and the maximum pseudo-energy is estimated to be $\Psi_{\text{max}} \approx 8.9$.

Next, we consider the Duffing oscillator with $m = 1$, $c = 1$, $k_1 = 1$, $k_3 = 20$, and $f^{\text{ext}}(t) = p_0 \cos(2\pi t)$ over the time duration of $[0, T]$. For this problem, we choose the load intensity $p_0 = 500$ and time duration $T = 1$ and initial conditions as $u_0 = 5$ and $v_0 = -10$. If we neglect the initial velocity and non-linear effects, then we can estimate the maximum displacement of the oscillator as $u_{\text{max}} \approx u_0 + 2p_0/k_1$. Thus, the maximum pseudo-energy stored in the system may be approximated as $\Psi_{\text{max}} \approx \frac{1}{2}mv_0^2 + \frac{1}{2}k_1u_{\text{max}}^2 \approx 5 \times 10^5$.

Reference solutions for both problems, the Van der Pol oscillator and Duffing oscillator, are obtained by solving them with the standard Bathe scheme^{29,30} and the 4th-order Runge–Kutta (RK4) scheme using a time-step of $\Delta t = 10^{-7}$ for both. The difference in the solutions obtained from these schemes is quantified using Equation (66) where $\mathbf{z}$ is taken to be the RK4 solution and $\mathbf{z}$ is taken to be the solution obtained from the Bathe scheme. The error (or difference) between these solutions is found to be of the order of 10^{-11} for the Van der Pol oscillator and of the order of 10^{-13} for the Duffing oscillator.

Next, both problems are solved using the proposed IMEX-BDFk-SAV schemes and other time integration methods such as the Bathe, TR, and generalized- α schemes. For implicit schemes such as the Bathe, TR and generalized- α , Newton–Raphson iterations are performed at each time-step until the absolute residual reaches a tolerance of 10^{-7} . Note that the proposed IMEX-BDFk-SAV schemes do not require iterations because non-linear terms are treated explicitly. Figures 4 and 5 show global errors in kinematic quantities using the proposed IMEX-BDFk-SAV schemes with $\psi = 100\Psi_{\text{max}}$, standard Bathe scheme, TR, and generalized- α scheme for the unforced Van der Pol oscillator and the forced Duffing oscillator, respectively. It can be observed that the proposed IMEX-BDFk-SAV schemes give k th-order accurate solutions for both non-linear problems. Lastly, note that errors for all time-steps considered are larger than about 10^{-9} for the Van der Pol oscillator and 10^{-11} for the Duffing oscillator, which are much larger than the difference between the Bathe and RK4 solutions that were used as a reference for these problems.

Figures 6 and 7 show the variation of global errors for the proposed IMEX-BDFk-SAV schemes for different values of ψ/Ψ_{max} . As with linear problems, the errors depend upon the choice of ψ , but for values of $\psi/\Psi_{\text{max}} > 10^2$, the errors stabilize. Thus, we suggest choosing a value of $\psi = 100\Psi_{\text{max}}$ for non-linear problems as well.

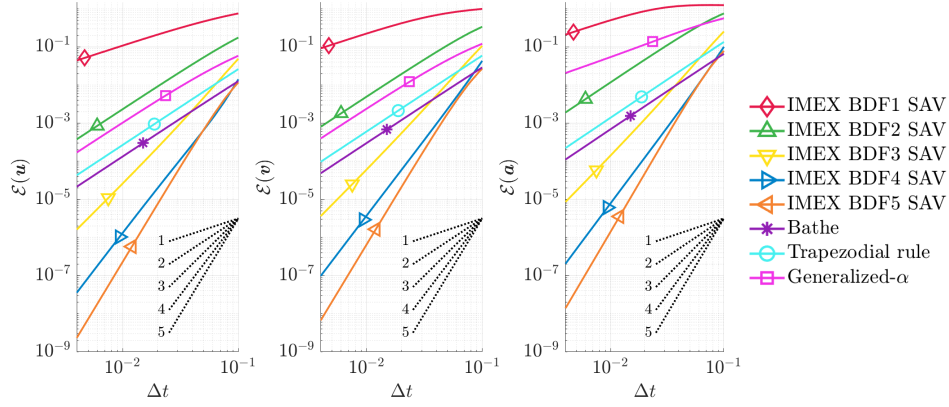


FIGURE 4 Error norms for the SDOF Van der Pol problem with $f^{\text{ext}}(t) = 0$, $u_0 = 2$, and $v_0 = 0$ for 29 different time-steps between 4×10^{-3} to 10^{-1} .

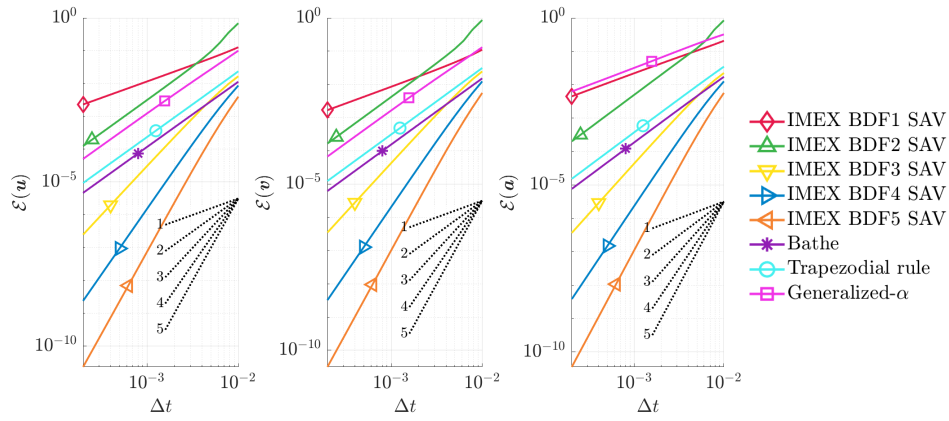


FIGURE 5 Error norms for the SDOF Duffing problem with $f^{\text{ext}}(t) = 500 \cos 2\pi t$, $u_0 = 5$, and $v_0 = -10$ for 18 different time-steps between 2×10^{-4} to 10^{-2} .

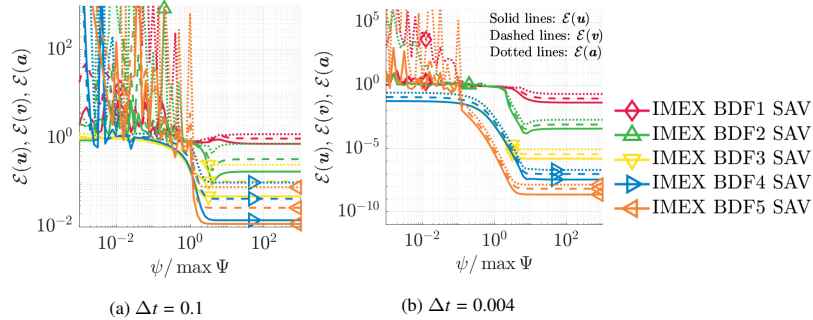


FIGURE 6 Error norms for the SDOF Van der Pol problem with $f^{\text{ext}}(t) = 0$, $u_0 = 2$, and $v_0 = 0$, using the proposed IMEX-BDF k -SAV schemes for various values of $\psi/\Psi_{\max} \in [10^{-3}, 10^3]$ at intervals of $10^{0.1}$.

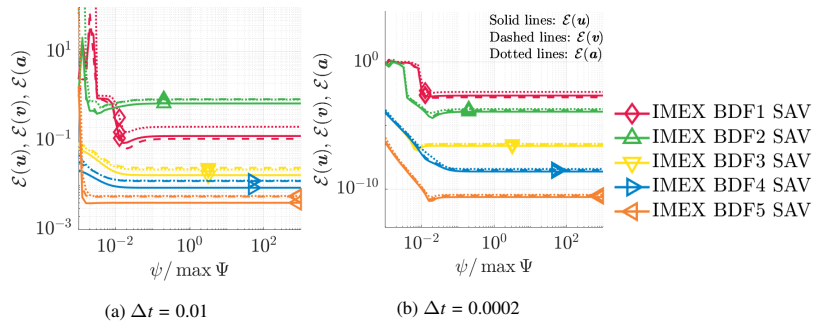


FIGURE 7 Error norms for the SDOF Duffing problem with $f^{\text{ext}}(t) = 500 \cos 2\pi t$, $u_0 = 5$, and $v_0 = -10$, using the proposed IMEX-BDF k -SAV schemes for various values of $\psi/\Psi_{\max} \in [10^{-3}, 10^3]$ at intervals of $10^{0.1}$.

Note that for the Van der Pol and Duffing oscillator problems, the internal force is decomposed into linear and non-linear parts following Equation (8). For the Van der Pol oscillator, $\mathbf{C} = 0$, $\mathbf{K} = k_1$, $\mathbf{f}^{NL} = -\mu(1 - u^2)\dot{u}$ and for the Duffing oscillator $\mathbf{C} = c$, $\mathbf{K} = k_1$, $\mathbf{f}^{NL} = k_3 u^3$. To investigate the effect of this decomposition of the internal force into linear and non-linear parts, we plot errors obtained for these problems for different values of $\mathbf{K}$ in Figures 8 and 9, while keeping the value of $\mathbf{C}$ constant (i.e. $\mathbf{C} = 0$ for the Van der Pol oscillator and $\mathbf{C} = c$ for the Duffing oscillator). We observe that the errors remain relatively uniform up to a certain threshold value of $\mathbf{K}/k_1$ and increase progressively for larger values. Thus, we suggest using the decomposition of the internal force as given in Equation (8) which corresponds to $\mathbf{K}/k_1 = 1$ in the examples above.

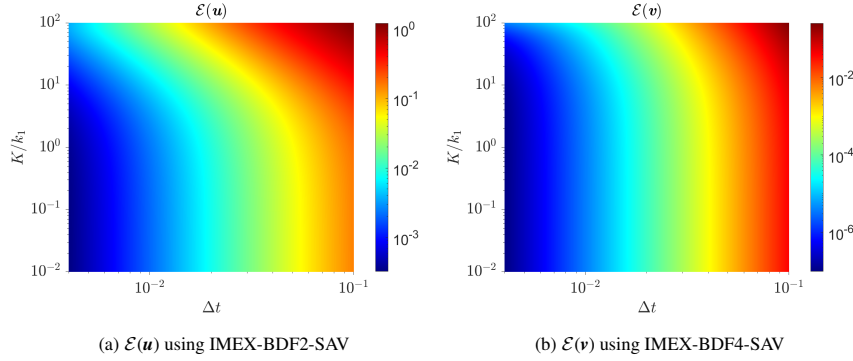


FIGURE 8 Error norms for the SDOF Van der Pol problem using the proposed IMEX-BDFk-SAV schemes for $\Delta t \in [4 \times 10^{-3}, 10^{-1}]$ and $\mathbf{K}/k_1 \in [10^{-2}, 10^2]$.

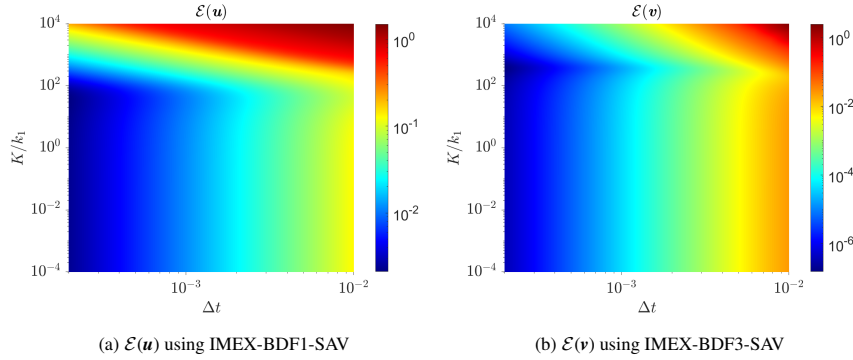


FIGURE 9 Error norms for the SDOF Duffing problem using the proposed IMEX-BDFk-SAV schemes for $\Delta t \in [2 \times 10^{-4}, 10^{-2}]$ and $\mathbf{K}/k_1 \in [10^{-4}, 10^4]$.

3 | NUMERICAL EXAMPLES

In this section, several numerical examples for non-linear structural dynamics are presented to demonstrate the performance of the proposed IMEX-BDFk-SAV schemes: (1) a simple pendulum model, (2) a spring-pendulum model, (3) a multi-degree-of-freedom (MDOF) Duffing oscillator, and (4) a 9-story building. Solutions obtained from the proposed IMEX-BDFk-SAV schemes with $\psi = 100\Psi_{\max}$, Bathe scheme^{29,30}, TR⁹, Generalized- α scheme¹³, Kim's 4th-order explicit scheme⁴², and central difference (CD) scheme are compared. Note that when non-linear internal forces depend on velocities, we employ the central difference method proposed by Park and Underwood⁷², to avoid the need for iterations at each time-step. To compare accuracy, we define three measures of error: the local instantaneous error ϵ_n^i , maximum instantaneous error ϵ_n , and maximum error ϵ :

$$\epsilon_n^i(\mathbf{z}) = |z_n^i - \tilde{z}_n^i| / \left(\max_{i,n} \tilde{z}_n^i - \min_{i,n} \tilde{z}_n^i \right), \quad \epsilon_n(\mathbf{z}) = \max_i \epsilon_n^i(\mathbf{z}), \quad \epsilon(\mathbf{z}) = \max_n \epsilon_n(\mathbf{z}) \quad (70)$$

where i denotes the degree of freedom ($i = 1, 2, \dots, N_i$), where N_i is the total number of degree of freedom. The variables z_n^i and $\bar{z}_n^i$ represent the numerical and reference solutions, respectively, at a degree of freedom i at time t_n .

3.1 | Simple Pendulum Model

We consider a simple pendulum – a benchmark problem for demonstrating the performance of non-linear systems^{34,35,73}. The equation of motion for this problem is given as:

$$\ddot{\theta}(t) + \frac{g}{L} \sin(\theta(t)) = 0 \quad (71)$$

$$\theta(t=0) = \theta_0, \quad \dot{\theta}(t=0) = v_0 \quad (72)$$

where θ , g , and L are the angle of swing, gravitational constant, and pendulum length, respectively. The exact solution to this problem can be obtained from the analytical relation between $\theta(t)$ and t ⁷³:

$$t = \pm \int_{\theta_0}^{\theta(t)} \left((v_0)^2 + 2 \frac{g}{L} (\cos z - \cos \theta_0) \right)^{-0.5} dz \quad (73)$$

Note that in the pendulum problem, we can modify the level of non-linearity by controlling θ_0 and v_0 . As θ_0 and/or v_0 increase, the system becomes more non-linear. Further details of this simple pendulum problem and comments on the importance of the problem are given in References^{35,73}.

The parameters chosen for this pendulum problem are $L = g = 1$, and initial conditions are $\theta_0 = 0$ and $v_0 = 1.95$ for which the maximum angle of swing of the pendulum computed from Equation (73), $\theta_{\max} = 2.6934$. The time duration of the simulation is $[0, 2T_1]$, where $T_1 = 11.6576$ is the period of system obtained from Equation (73). During the simulation, the maximum pseudo-energy is approximated as $\Psi_{\max} \approx \max(\frac{1}{2}v_0^2, \frac{1}{2}\frac{g}{L}\theta_{\max}^2) \approx 3.6$ because the system is conservative. Thus, we choose $\psi = 100\Psi_{\max} = 360$ for the proposed IMEX-BDFk-SAV schemes.

Figure 10 shows numerical results for the angle of swing for different time integration schemes. The time-step Δt for the proposed IMEX-BDFk-SAV schemes is chosen as $T_1/20$ and $T_1/100$. For computational cost parity, the time-steps of the comparative schemes are chosen as $n_{\text{sub}} \times \Delta t$, where n_{sub} is the number of sub-stages associated with the time integration scheme. For instance, n_{sub} for the Bathe scheme is 2 and for the Kim's 4th-order explicit scheme n_{sub} is 4. Note that for the TR scheme, generalized- α scheme, and CD scheme, number of sub-stages $n_{\text{sub}} = 1$. For sake of clarity, we do not include the conventional IMEX-BDFk schemes (without SAV) in this figure because their response is almost identical to that of the corresponding IMEX-BDFk-SAV schemes for this problem. We observe that when $\Delta t = T_1/20$, several schemes drift away from the exact solution because when $|\theta|$ exceeds π then the pendulum makes more than a half-revolution and consequently cannot maintain oscillations around the equilibrium position. With a finer time-step of $\Delta t = T_1/100$, none of the methods suffer from this problem, however, IMEX-BDF1 and IMEX-BDF1-SAV lead to large numerical damping and large period shortening, and IMEX-BDF2 and IMEX-BDF2-SAV have an uncharacteristically large period elongation.

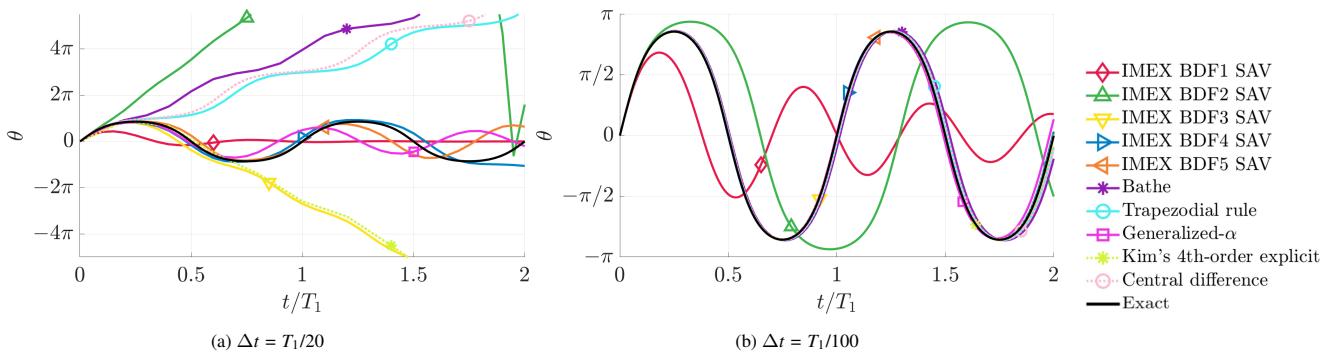


FIGURE 10 Time history of angles for various schemes in the simple pendulum model.

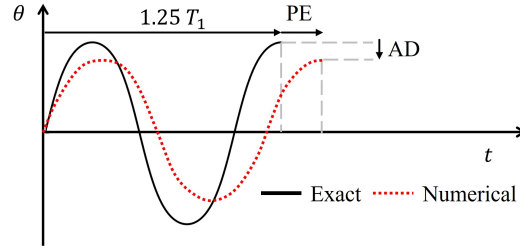


FIGURE 11 Definition of period elongation (PE) and amplitude decay (AD) in the simple pendulum model.

TABLE 1 Period elongation (PE), amplitude decay (AD), and computational costs for various schemes over the time interval $[0, 2T_1]$ in the simple pendulum problem.

Scheme	Δt	Average number of iterations		Computational cost (μs)		$PE/1.25T_1$ (%)		$AD/\theta_{\max}$ (%)	
		$T_1/20$	$T_1/100$	$T_1/20$	$T_1/100$	$T_1/20$	$T_1/100$	$T_1/20$	$T_1/100$
IMEX-BDF1-SAV	1	1	1	14	65	-38.5528	-32.3366	91.9572	53.1368
IMEX-BDF2-SAV	1	1	1	19	80	*	28.4552	*	-8.8956
IMEX-BDF3-SAV	1	1	1	25	120	*	0.3494	*	-0.4102
IMEX-BDF4-SAV	1	1	1	25	122	-1.6876	-0.2106	-8.4120	0.1253
IMEX-BDF5-SAV	1	1	1	25	129	-5.8110	0.0344	10.6965	-0.0224
Bathe	2	1.115	1.115	140	345	*	1.0886	*	-0.6543
TR	2	1.045	1.045	146	356	*	0.7056	*	-0.3551
Generalized- α	2.35	1.555	1.555	150	408	-13.8742	-0.4199	29.1999	0.7268
Kim's 4 th -order	1	1	1	13	79	*	-0.6060	*	0.0318
CD	1	1	1	45	176	*	0.7561	*	-0.3831

* The pendulum cannot maintain oscillations around the equilibrium position.

To compare the accuracy of the different schemes, we measure errors in terms of period elongation and amplitude decay using the approach presented by Bathe and Wilson⁷⁴. These measures can be obtained by comparing the exact solution in Equation (73) with the numerical solutions (see Figure 11). In Table 1, we summarize the average number of iterations per time-step, computational cost, percentage period elongation ($PE/1.25T_1$), and amplitude decay ($AD/\theta_{\max}$) for several time integration schemes. We observe that the computational cost of the proposed IMEX-BDFk-SAV schemes is at par or lower than existing explicit methods with the added benefit of unconditional stability over the explicit methods. Further, at the smaller time-step ($\Delta t = T_1/100$), period elongation and amplitude decay improve with the order of the proposed IMEX-BDFk-SAV schemes for $k = 3, 4$, and 5 . This trend is found to continue for progressively smaller time-steps as well, which are not shown in this table. This numerical example demonstrates the main benefits of the IMEX-BDFk-SAV schemes, i.e., unconditional stability, low computational cost, no need for iterations for non-linear problems, and high-order accuracy.

3.2 | Spring-Pendulum Model

We solve an elastic spring-pendulum problem^{14,35,36} as shown in Figure 12. The governing differential equation of the problem can be expressed as:

$$\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} + \underbrace{\begin{bmatrix} k/m & 0 \\ 0 & g/L_0 \end{bmatrix}}_{\text{Linear stiffness}} \begin{bmatrix} x \\ \theta \end{bmatrix} + \underbrace{\begin{bmatrix} -(L_0 + x)(\dot{\theta})^2 - g \cos \theta \\ (2\dot{x}\dot{\theta} + g \sin \theta)/(L_0 + x) - g\theta/L_0 \end{bmatrix}}_{\text{Non-linear force}} = \begin{bmatrix} f_x \\ f_\theta \end{bmatrix} \quad (74)$$

where x and θ denote the radial and circumferential displacements, respectively, m is the mass of the pendulum, k is the spring constant, L_0 is the initial length of the pendulum, and g is the gravitational constant. The parameters for this problem are selected as $m = 1$, $k = 98.1$, $L_0 = 0.5$, and $g = 9.81$. In this study, we consider a manufactured analytical solution in time from $t = 0$ to $t = 2$:

$$x(t) = \theta(t) = 0.1 \sin(2\pi t) \quad (75)$$

Substituting Equation (75) into Equation (74), the required forcing functions f_x and f_θ are obtained. For this non-linear system, the maximum pseudo-energy is $\Psi_{\max} = 0.59$. Thus, for the proposed IMEX-BDFk-SAV schemes, we select $\psi = 100\Psi_{\max} = 59$.

Figure 13 shows time histories of displacements for various time integration schemes with $\Delta t = T_1/10$ and $T_1/20$ for the proposed IMEX-BDFk-SAV schemes. Similar to previous example, the time-steps of the comparative schemes are selected as

$n_{\text{sub}} \times \Delta t$. In this problem, since the internal force depends upon velocities, we employ the central difference method proposed by Park and Underwood⁷², i.e. CD (PU), to avoid the need for iterations at each time-step. Note that, at the larger time-step ($n_{\text{sub}} \times \Delta t = 4T_1/10$), Kim's 4th-order scheme is unstable. We also observe that the IMEX-BDF1-SAV scheme, being a low-order scheme, has high errors for both the time-steps considered here. The performance of different time integration schemes for this problem is summarized in Table 2. We note that, at the smaller time-step, the IMEX-BDF4-SAV and IMEX-BDF5-SAV schemes lead to the lowest error, have low computational cost and are guaranteed to be unconditionally stable.

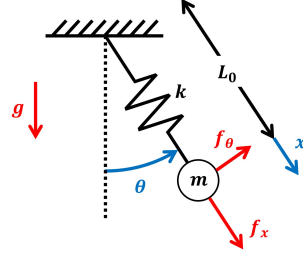


FIGURE 12 A 2DOF spring-pendulum model.

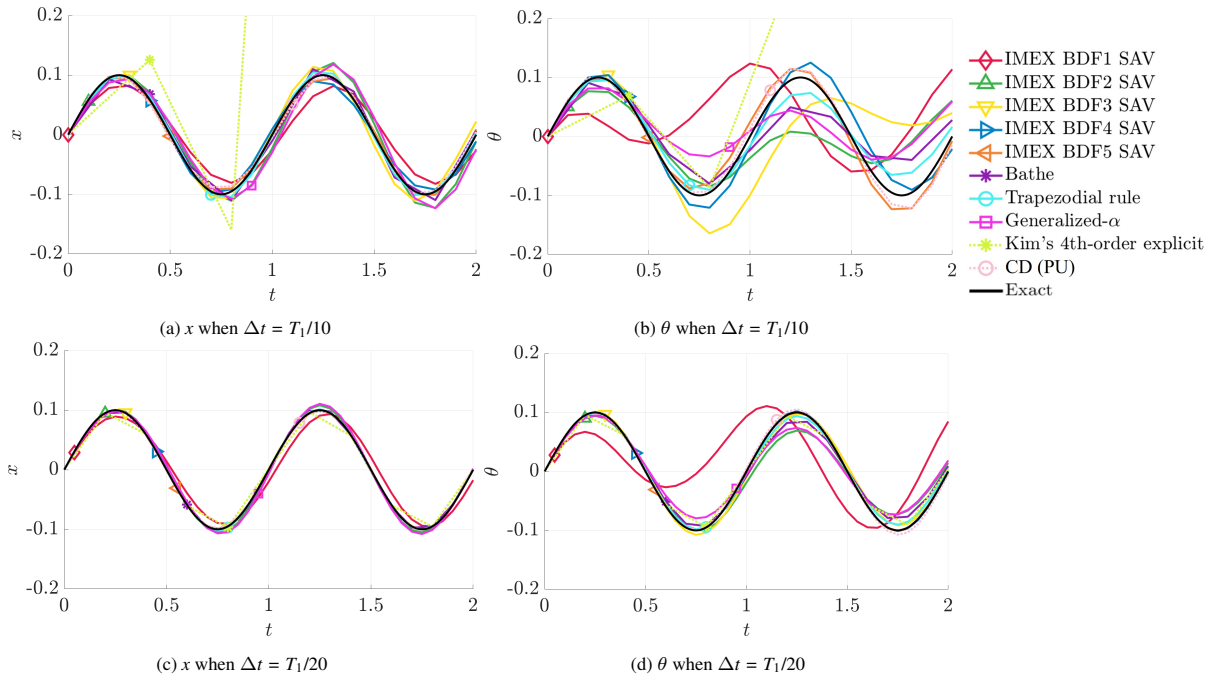


FIGURE 13 Time history of the radial displacement x and circumferential displacement θ .

3.3 | Multi-Degree-Of-Freedom (MDOF) Duffing Oscillator

In this section, we solve a 20-DOF Duffing oscillator system as shown in Figure 14. The Duffing oscillators are connected in series such that DOFs i and $i - 1$ are connected with a damper with coefficient c_i , linear hardening stiffness $k_{1,i}$, and cubic hardening stiffness $k_{3,i}$. Conceptually, this problem can be thought of as a 20-story shear building with non-linear springs. The

TABLE 2 Maximum errors and computational cost for various schemes using $\Delta t = T_1/10$ and $T_1/20$ over the time interval $[0, 2T_1]$ in the spring-pendulum problem.

Scheme	Δt	Average number of iterations		Computational cost (μs)		Error (%)					
		$T_1/10$	$T_1/20$	$T_1/10$	$T_1/20$	$\epsilon(u)$		$\epsilon(v)$		$\epsilon(a)$	
		$T_1/10$	$T_1/20$	$T_1/10$	$T_1/20$	$T_1/10$	$T_1/20$	$T_1/10$	$T_1/20$	$T_1/10$	$T_1/20$
IMEX-BDF1-SAV	1	1	1	178	326	79.79	56.56	51.40	35.70	43.57	29.93
IMEX-BDF2-SAV	1	1	1	167	335	45.18	15.47	33.90	11.32	37.13	9.89
IMEX-BDF3-SAV	1	1	1	203	323	59.66	5.32	38.03	3.84	34.60	2.99
IMEX-BDF4-SAV	1	1	1	176	306	20.96	0.98	13.07	0.67	13.21	0.61
IMEX-BDF5-SAV	1	1	1	157	301	14.17	0.40	9.38	0.28	19.00	0.22
Bathe	2.95	2.425		409	743	27.61	9.14	18.38	5.59	19.00	5.74
TR	2.8	2		418	622	16.84	4.92	10.74	3.00	11.30	3.02
Generalized- α	3	2.5		613	1012	41.15	15.20	31.70	11.52	48.35	23.00
Kim's 4 th -order	1	1		*	105	*	2.27	*	1.51	*	2.30
CD (PU)	1	1		94	182	13.32	4.25	14.29	6.18	10.80	4.22

* Divergence

equation of motion at DOF i of this system can be written as:

$$m_i \ddot{u}^i + \underbrace{c_i(\dot{u}^i - \dot{u}^{i-1}) + c_{i+1}(\dot{u}^i - \dot{u}^{i+1})}_{\text{Linear damping}} + \underbrace{k_{1,i}(u^i - u^{i-1}) + k_{1,i+1}(u^i - u^{i+1})}_{\text{Linear stiffness}} + \underbrace{k_{3,i}(u^i - u^{i-1})^3 + k_{3,i+1}(u^i - u^{i+1})^3}_{\text{Non-linear force}} = f^{\text{ext},i} \quad (76)$$

We select the following values of the problem parameters: $m_i = 1$, $c_i = 0.3$, $k_{1,i} = 1$ and $k_{3,i} = 10$ for all $i = 1, 2, \dots, 20$. External force, $f^{\text{ext},i} = 0$ for $i = 1, 2, \dots, 19$, and $f^{\text{ext},20} = \cos \omega_p t$, where $\omega_p = 1$. We impose zero initial displacement and velocity, $\mathbf{u}(0) = \mathbf{v}(0) = \mathbf{0}$. During the simulation, the maximum value of pseudo-energy stored in the underlying undamped linear system is approximated as $\Psi_{\max} \approx \frac{1}{2} \mathbf{u}_{\max}^T \mathbf{K} \mathbf{u}_{\max} \approx 40$, where $\mathbf{u}_{\max} \approx 2[\mathbf{K}]^{-1} \max(\mathbf{f}^{\text{ext}})$, $\mathbf{K}$ is the linear stiffness matrix, and $\mathbf{f}^{\text{ext}}$ is external force vector. Thus, we choose $\psi = 100\Psi_{\max} = 4000$ for the proposed IMEX-BDFk-SAV schemes.

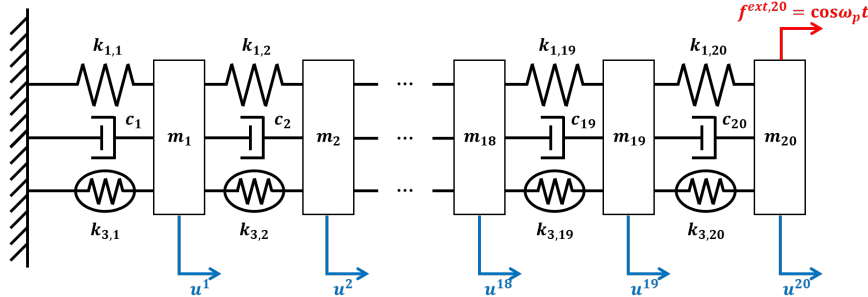


FIGURE 14 A non-linear system comprising 20 Duffing oscillators.

To generate a reference solution to this problem, the standard Bathe scheme^{29,30}, Kim's 4th-order scheme, and the RK4 method are used to solve this problem with a time-step of 10^{-5} . The maximum difference (error), as defined in Equation (70), between the solutions obtained from these methods is found to be of the order of 10^{-10} . With this difference (error) in mind, the standard Bathe scheme with the time-step size of 10^{-5} is taken to be a reference solution for comparing errors among different time integration schemes for this problem.

Figure 15 shows time histories of all kinematic quantities at node 10 over the time duration $[0, 50]$ for the proposed IMEX-BDFk-SAV, Bathe, TR, and generalized- α , Kim's 4th-order, and CD schemes. The time-step chosen for this problem is $n_{\text{sub}} \times \Delta t$, where $\Delta t = 0.2$. Note that Kim's 4th-order scheme is unstable at this large time-step whereas the proposed IMEX-BDFk-SAV schemes are always unconditionally stable and they achieve up to 5th-order accuracy.

Figure 16 shows the comparison between maximum errors and computational cost for different time integration schemes when Δt is varied from 0.005 to 0.16 by doubling it successively 5 times. Note that, for the same computational cost, explicit CD and Kim's 4th-order schemes yield more accurate results than the proposed IMEX-BDF2-SAV and IMEX-BDF4-SAV schemes, respectively. However, explicit schemes are only conditionally stable and require a small time-step to compute stable solutions.

On the other hand, compared to 2nd-order implicit schemes, we note that the IMEX-BDF2-SAV scheme offers the best trade-off between computational cost and accuracy. Thus, with the proposed IMEX-BDF k -SAV schemes, one essentially gets the benefit of unconditional stability, high accuracy, and the lowest computational cost.

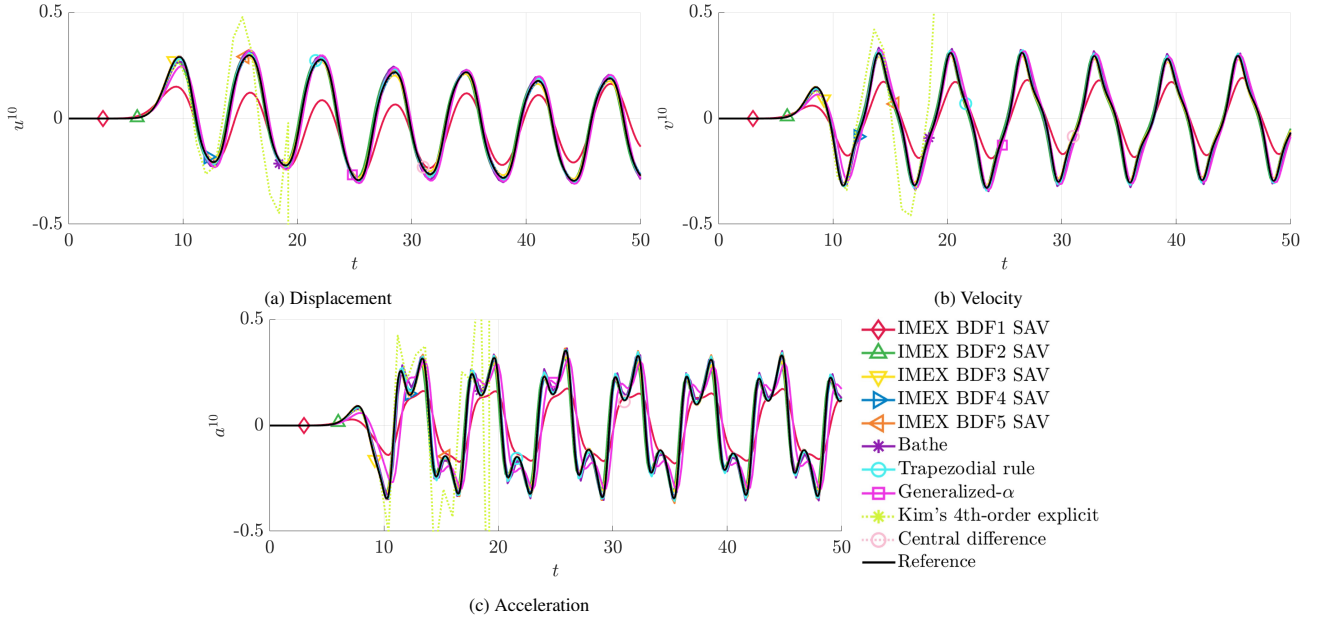


FIGURE 15 Time histories of displacement, velocity, and acceleration at node 10 when $\Delta t = 0.2$.

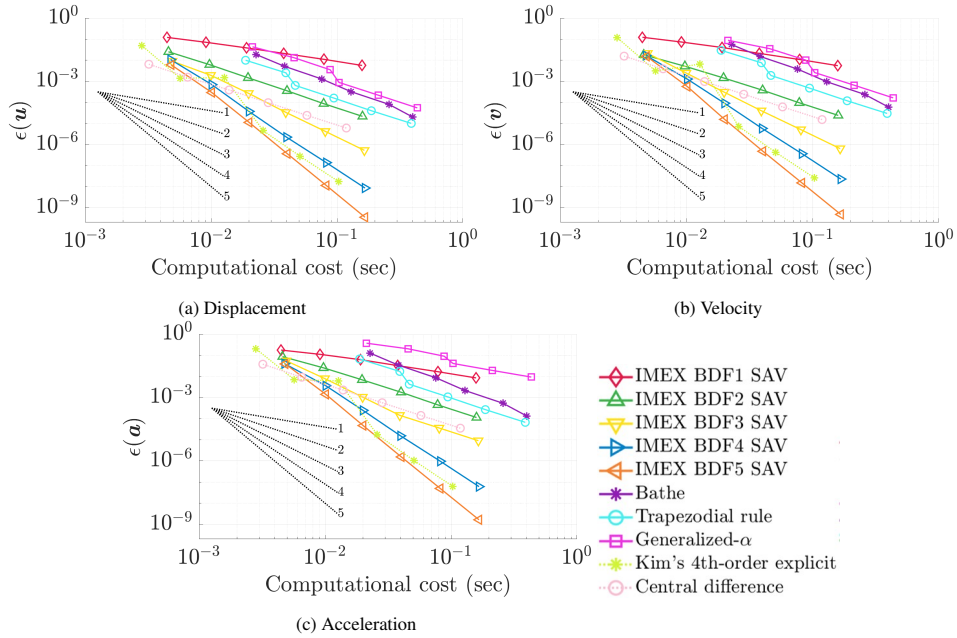


FIGURE 16 Maximum errors in displacement, velocity, and acceleration. The 6 points on each curve correspond to values of $\Delta t \in \{0.005, 0.01, 0.02, 0.04, 0.08, 0.16\}$.

3.4 | Nine-Story Building Subject to El Centro Earthquake

As shown in Figure 17, we consider a 9-story building with a Buckling-Restrained Axial Damper (BRAD) on the first floor. This 9-story structure has been used as a benchmark problem to study seismically-excited buildings in^{75–77}. The geometry, material properties, and boundary conditions are identical to those of the nine-story benchmark building north-south (N-S) moment-resisting frame in Ohtori et al.⁷⁵. Initial displacement and velocity vectors are taken to be zero and the following external force is applied to the structure:

$$\mathbf{f}^{\text{ext}}(t) = -\mathbf{M}\mathbf{\Gamma}\ddot{x}_g(t) \quad (77)$$

where $\ddot{x}_g$ is the ground acceleration and $\mathbf{\Gamma}$ is a loading vector that describes the inertial effects due to the ground acceleration. The input ground motion is selected as the N-S component recorded at the Imperial Valley Irrigation District substation in El Centro, California, during the Imperial Valley, California earthquake of May 18, 1940 (see Dyke⁷⁸). The input ground motion in Dyke⁷⁸ is recorded at uniformly spaced intervals of 0.02 s. Thus, when a time-instant of t_i is not a multiple of 0.02 sec, the ground acceleration $\ddot{x}_g(t_i)$ is obtained by shape-preserving piecewise cubic interpolation.

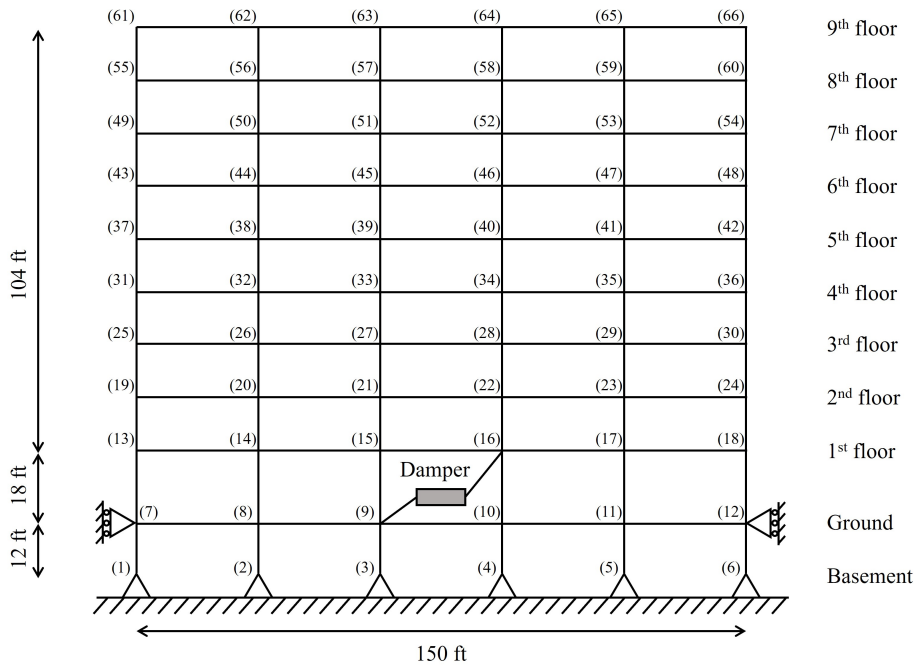


FIGURE 17 A 9-story building with a Buckling-Restrained Axial Damper (BRAD).

We use one 2-noded Euler-Bernoulli frame element within each column and beam for spatial discretization. The mass matrix is obtained using the HRZ lumping scheme⁷⁹. The corresponding first five natural frequencies are 0.443, 1.16, 1.99, 2.92, and 3.94 Hz, which are consistent with those in Ohtori et al.⁷⁵. Mass- and stiffness-proportional Rayleigh damping coefficients are selected such that the first and fifth modal damping ratios are 0.02: $\zeta_1 = \zeta_5 = 0.02$. For modeling the BRAD, we use the extended Bouc–Wen model⁸⁰, where the restoring force $F(t)$ and hysteretic displacement $z(t)$ are given as:

$$F(t) = \alpha k u(t) + (1 - \alpha) k z(t), \quad \dot{z}(t) = \dot{u}(t) [A - \beta |z(t)|^p - \gamma |z(t)|^q \text{sgn}(z(t)) \dot{y}(t)] \quad (78)$$

where sgn denotes the sign function, and α , k , A , β , γ , p , and q are model parameters. In this study, the following model parameters are chosen: $A = 1$, $\alpha = 0.015$, $p = 4.6$, $q = 0.925$, $k = \varphi F_u / u_u$, $\beta = \hat{\beta} / |u_u|^p$, and $\gamma = \hat{\gamma} / |u_u|^q$ where $u_u = 0.02$ m, $F_u = 2 \times 10^5$ N, $\varphi = 7.756$, $\hat{\beta} = -9.856$, and $\hat{\gamma} = 7.185$ (see Sireteanu, Giuclea and Mitu⁸⁰).

A reference solution is obtained by solving the problem with the RK4 scheme and with Kim's 4th-order scheme using a small time-step of 10^{-6} sec. The error (or difference) between these two solutions is found to be of the order of 10^{-13} in displacement. The solution from the RK4 scheme is taken as the reference solution for the computation of errors in the following discussion.

In the absence of linear and non-linear damping, we estimate the maximum pseudo-energy as $\Psi_{\max} \approx \frac{1}{2} \mathbf{u}_{\max}^T \mathbf{K} \mathbf{u}_{\max} \approx 12$ MJ, where $\mathbf{u}_{\max} \approx 2[\mathbf{K}]^{-1} \max(|\mathbf{f}^{\text{ext}}|)$. Similar to other examples, ψ for the proposed IMEX-BDF k -SAV schemes is selected as $100\Psi_{\max} = 1.2$ GJ. For certain IMEX-BDF k -SAV schemes, however, we observe *temporal locking* as described in Section 2.2. For example, Figure 18 shows the numerical solution obtained using the IMEX-BDF1 and IMEX-BDF1-SAV schemes with and without the SAV recovery. Note that around $t = 2.1$ sec, the underlying IMEX-BDF1 solution diverges. At the same time, IMEX-BDF1-SAV solution without SAV recovery locks ≈ 0 . The reason for this temporal locking, as explained in Section 2.2, is that Φ_{n+1} becomes ≈ 0 at all times after this instant as shown in Figure 18b. In contrast, with the SAV recovery algorithm, Φ_{n+1} maintains its value of around 1.2 GJ, and temporal locking is avoided.

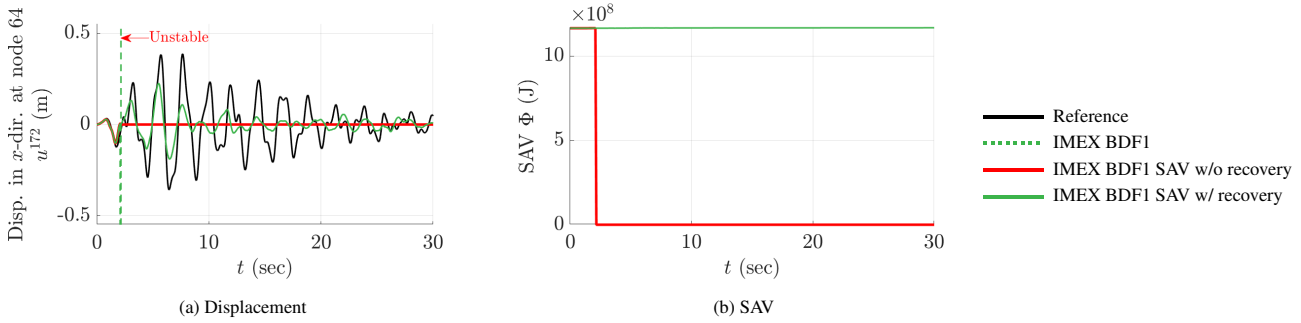


FIGURE 18 Time history of SAV and displacement obtained from IMEX-BDF1-SAV schemes with and without the proposed recovery algorithm for $\Delta t = 0.08$ sec.

In Figure 19, sub-figure (a) presents the time history of displacement in the x -direction at node 64 (DOF 172), sub-figure (b) shows the time-history of error, and sub-figure (c) shows a comparison of the computational cost and accuracy of the different time integration schemes. In Figure 19a,b, the time-steps for the various schemes presented are chosen such that the computational cost of all the schemes is similar – around 3.5 sec, as highlighted in Figure 19c. The specific time-steps chosen and associated computational costs of all the schemes considered are listed in Table 3. Figure 19b shows that for all schemes, the local instantaneous error is higher in the initial part of the simulation when the earthquake excitation is higher and it steadily decreases with time. Figure 19c shows that the conventional IMEX-BDF5 and the central difference schemes are unstable for time-steps $\Delta t \geq 0.0032$ sec and Kim's 4th-order explicit scheme is unstable for time-step $(n_{\text{sub}} \times \Delta t) \geq (n_{\text{sub}} \times 0.0004 \text{ sec})$ (where $n_{\text{sub}} = 4$). Further, from Table 3, we note that the proposed IMEX-BDF k -SAV schemes give stable and progressively accurate solutions as k increases from 1 to 5. One exception to this statement is that the IMEX-BDF5-SAV scheme results in poor accuracy for $\Delta t \geq 0.0032$ sec. Note that for this time-step, the underlying IMEX-BDF5 scheme is actually unstable. Consequently, the associated IMEX-BDF k -SAV scheme without SAV recovery exhibits temporal locking. With recovery, however, it results in poor accuracy for some duration of the simulation, but it remains unconditionally stable. Note that this problem has a non-zero forcing function, allowing the proposed IMEX-BDF k -SAV schemes to recover from the local temporal inaccuracy and give stable and accurate solutions for the remainder of the simulation. For example, Figure 20 shows that for $\Delta t = 0.02$ sec (a relatively large time-step for this problem), the solution of the IMEX-BDF2 scheme diverges about 5 sec into the simulation. At this point, we see a sharp rise in error of the corresponding IMEX-BDF2-SAV scheme which steadily decreases over time. Finally, we observe that the proposed IMEX-BDF k -SAV schemes with $k \geq 2$ provide the most accurate results when compared to other implicit and explicit schemes for the same computational cost.

Lastly, to demonstrate the performance of the proposed IMEX-BDF k -SAV schemes for elasto-plastic dynamics, we replace the Bouc–Wen model for the damper with an equivalent J2 plasticity-based material model⁶⁹ with the following material parameters: elastic modulus $E = 46$ MPa, yields stress $\sigma_Y = 185$ kPa, isotropic hardening modulus $K = 2$ kPa, and kinematic hardening modulus $H = 1.2$ MPa. Figure 21 shows a comparison of the hysteresis of the two damper models obtained using the IMEX-BDF3-SAV scheme with $\Delta t = 0.0008$ sec. This and other numerical examples presented show that the proposed IMEX-BDF k -SAV schemes work well for a variety of non-linear problems in structural dynamics.

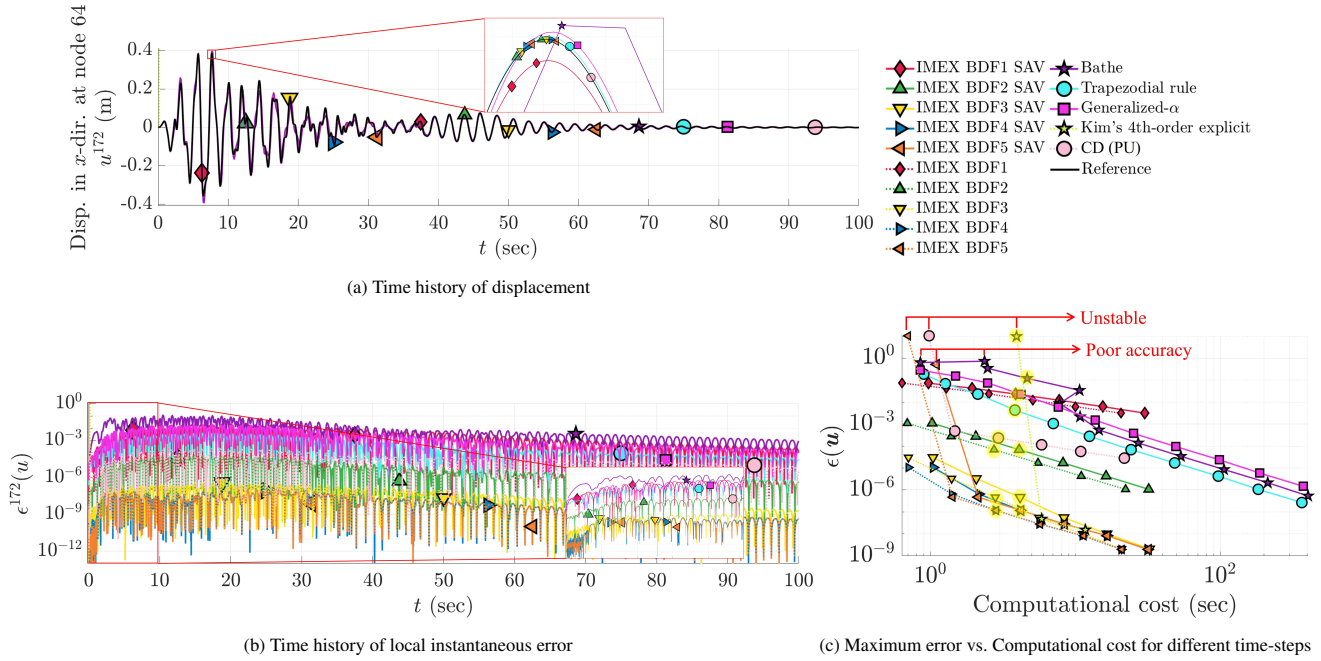


FIGURE 19 (a) Time history of displacement in x -direction at node 64 for different time-steps (corresponding points in sub-figure (c) are highlighted), (b) its local instantaneous error $\epsilon^{172}(d)$, (c) maximum error in displacement at any DOF for different time integration schemes. In sub-figure (c), the smallest time-step for each scheme considered is $n_{\text{sub}} \times \Delta t$ where $\Delta t = 0.0001$ sec and the remaining points on each curve are obtained by successively doubling this time-step. Note that the conventional IMEX-BDF5 and the central difference schemes are unstable for time-steps $\Delta t \geq 0.0032$ sec and Kim's 4th-order explicit scheme is unstable for time-step $(4 \times \Delta t) \geq (4 \times 0.0004)$ sec.

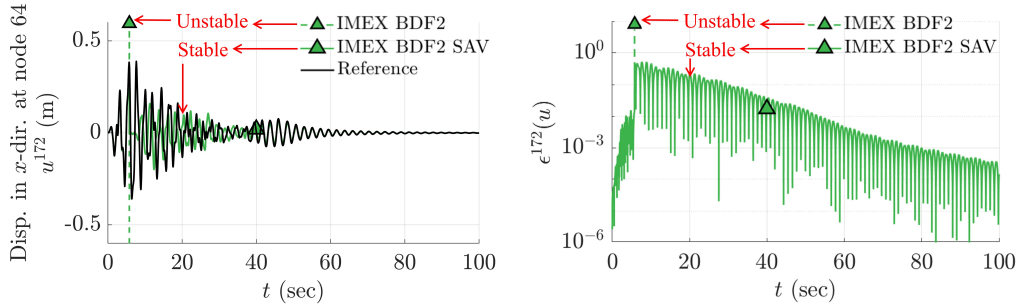


FIGURE 20 Time history of displacement and local instantaneous error obtained from IMEX-BDF2 and IMEX-BDF2-SAV schemes for $\Delta t = 0.02$ sec. Note the unconditional stability of IMEX-BDF2-SAV scheme when the underlying IMEX-BDF2 scheme is unstable.

TABLE 3 Summary of time-steps, computational costs and maximum errors for different time-integration schemes presented in Figure 19

Scheme	Δt (sec)	Computational cost (sec)	Error $\epsilon(u)$	Scheme	Δt (sec)	Computational cost (sec)	Error $\epsilon(u)$
IMEX-BDF1-SAV	0.0008	3.867389	2.3×10^{-2}	Bathe	0.0512	4.639049	1.2×10^{-1}
IMEX-BDF2-SAV	0.0008	4.086772	6.8×10^{-5}	TR	0.0128	3.823075	4.3×10^{-3}
IMEX-BDF3-SAV	0.0008	4.147435	4.6×10^{-7}	Generalized- α	0.0128	4.202484	2.2×10^{-2}
IMEX-BDF4-SAV	0.0008	4.166472	1.2×10^{-7}	Kim's 4 th explicit	0.0004	*	*
IMEX-BDF5-SAV	0.0008	4.208458	1.2×10^{-7}	CD (PU)	0.0008	2.936435	2.3×10^{-4}
IMEX-BDF1	0.0008	2.532614	2.3×10^{-2}				
IMEX-BDF2	0.0008	2.741602	6.8×10^{-5}				
IMEX-BDF3	0.0008	2.797186	4.6×10^{-7}				
IMEX-BDF4	0.0008	2.806063	1.2×10^{-7}				
IMEX-BDF5	0.0008	2.842503	1.2×10^{-7}				

* Divergence

4 | CONCLUDING REMARKS

In this study, we present a SAV stabilization of IMEX BDF k schemes for structural dynamics, and its stability and error analysis. Unlike existing SAV schemes, SAV stabilization is designed to solve second-order ODEs, such as general non-linear problems

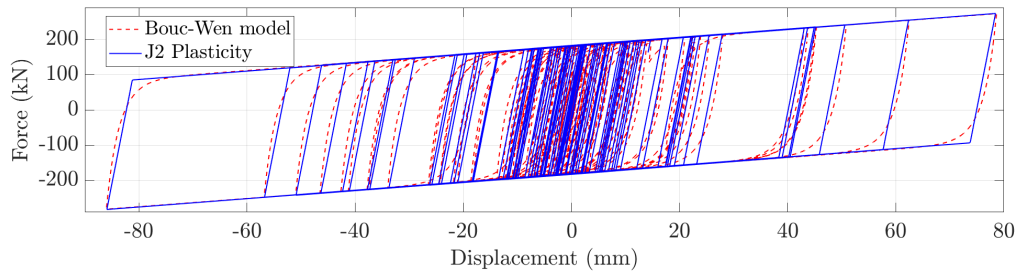


FIGURE 21 Comparison of the hysteretic response of the damper obtained using the Bouc-Wen model and the J2 plasticity model with the IMEX-BDF3-SAV scheme and $\Delta t = 0.0008$ sec.

in structural dynamics. In the proposed IMEX-BDF k -SAV schemes, linear terms are handled implicitly while non-linear terms are handled explicitly. Thus, there is no need for iterative algorithms, such as Picard iteration and Newton–Raphson iteration, to solve non-linear terms. Unlike conventional IMEX methods, which are conditionally stable, the proposed IMEX-BDF k -SAV schemes employ a novel scalar auxiliary variable to maintain unconditional stability. In addition, the proposed schemes are able to achieve k th-order accuracy.

A rigorous stability analysis is conducted to show that the proposed schemes are unconditionally stable for both linear and non-linear dynamics. Moreover, we carry out a detailed truncation error analysis. From the error analysis, we show that the optimal value of parameter $\beta^{(k)}$, given by Equation (22), is about half of its value in existing IMEX-BDF k -SAV schemes for other dynamical systems governed by first-order ODEs. This leads to a lower energy bound for unconditional stability. We also prove that the proposed IMEX-BDF k -SAV schemes with these values preserve the local truncation errors of the conventional BDF k schemes for linear structural dynamics. Convergence tests are conducted to validate these theoretical findings.

We illustrate the performance of the proposed IMEX-BDF k -SAV schemes by solving various non-linear problems: a simple pendulum, a spring-pendulum, a MDOF Duffing oscillator, a 9-story building with a non-linear damper subjected to an earthquake excitation. We compare the numerical results obtained using the proposed schemes and several implicit (Bathe, TR, and generalized- α) and explicit (Kim’s 4th-order and CD) time integration schemes using different measures of error. These numerical examples demonstrate that the proposed IMEX-BDF k -SAV schemes yield accurate solutions at a low computational cost while maintaining unconditional stability.

ACKNOWLEDGMENTS

This work is partly supported by the National Science Foundation (NSF) through Grant CNS-2229136: Collaborative Research: CPS: Co-Designed Control and Scheduling Adaptation for Assured Cyber-Physical System Safety and Performance. The authors also acknowledge support from the Computational Interdisciplinary Graduate Program (CIGP) of Purdue University.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Kwon SB, Prakash A. Scalar auxiliary variable (SAV) stabilization of implicit-explicit (IMEX) time integration schemes for nonlinear structural dynamics. *Int J Numer Methods Engng* 2024;00(00):1–26.

APPENDIX

A LOCAL TRUNCATION ERRORS IN FORCED RESPONSES

With initial displacement u_0 and velocity v_0 given, the LTEs of the proposed IMEX-BDFk-SAV schemes for forced responses (i.e. $f^{\text{ext}} = p_0 \sin(\omega_f t)$) are obtained as follows. Note that LTEs in displacements and velocities are found to be of the order of Δt^{k+1} : $\tau_u \propto O(\Delta t^{k+1})$ and $\tau_v \propto O(\Delta t^{k+1})$ for $1 \leq k \leq 5$:

i) $k = 1$:

$$\begin{bmatrix} \tau_u \\ \tau_v \end{bmatrix} \propto -\frac{\omega_0^2 \Delta t^2}{2} \begin{bmatrix} u_0(1 + 2\zeta\kappa) \\ v_0(1 - 4\zeta^2 - 2\zeta/\kappa) \end{bmatrix} + \frac{p_0 \omega_f \Delta t^2}{2} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + O(\Delta t^3) \quad (\text{A1})$$

ii) $k = 2$:

$$\begin{bmatrix} \tau_u \\ \tau_v \end{bmatrix} \propto -\frac{2\omega_0^3 \Delta t^3}{9} \begin{bmatrix} u_0(-2\zeta + (1 - 4\zeta^2)\kappa) \\ v_0(-4\zeta(1 - 2\zeta^2) - (1 - 4\zeta^2)/\kappa) \end{bmatrix} + \frac{2p_0 \omega_f \Delta t^3}{9} \begin{bmatrix} 1 \\ -2\omega_0 \zeta \end{bmatrix} + O(\Delta t^4) \quad (\text{A2})$$

iii) $k = 3$:

$$\begin{bmatrix} \tau_u \\ \tau_v \end{bmatrix} \propto \frac{3\omega_0^4 \Delta t^4}{22} \begin{bmatrix} u_0(1 - 4\zeta^2 + 4\zeta(1 - 2\zeta^2)\kappa) \\ v_0(1 - 12\zeta^2 + 16\zeta^4 - 4\zeta(1 - 2\zeta^2)/\kappa) \end{bmatrix} - \frac{3p_0\omega_f \Delta t^4}{22} \begin{bmatrix} 2\omega_0\zeta \\ \omega_0^2(1 - 4\zeta^2) + \omega_f^2 \end{bmatrix} + O(\Delta t^5) \quad (\text{A3})$$

iv) $k = 4$:

$$\begin{bmatrix} \tau_u \\ \tau_v \end{bmatrix} \propto \frac{12\omega_0^5 \Delta t^5}{125} \begin{bmatrix} u_0(-4\zeta(1 - 2\zeta^2) + (1 - 12\zeta^2 + 16\zeta^4)\kappa) \\ v_0(-2\zeta(3 - 16\zeta^2 + 16\zeta^4) - (1 - 12\zeta^2 + 16\zeta^4)/\kappa) \end{bmatrix} - \frac{12p_0\omega_f \Delta t^5}{125} \begin{bmatrix} \omega_0^2(1 - 4\zeta^2) + \omega_f^2 \\ -2\omega_0\zeta(2\omega_0^2(1 - 2\zeta^2) + \omega_f^2) \end{bmatrix} + O(\Delta t^6) \quad (\text{A4})$$

v) $k = 5$:

$$\begin{aligned} \begin{bmatrix} \tau_u \\ \tau_v \end{bmatrix} &\propto -\frac{10\omega_0^6 \Delta t^6}{137} \begin{bmatrix} u_0(1 - 12\zeta^2 + 16\zeta^4 + 2\zeta(3 - 16\zeta^2 + 16\zeta^4)\kappa) \\ v_0(1 - 24\zeta^2 + 80\zeta^4 - 64\zeta^6 - 2\zeta(3 - 16\zeta^2 + 16\zeta^4)/\kappa) \end{bmatrix} \\ &+ \frac{10p_0\omega_f \Delta t^6}{137} \begin{bmatrix} -2\omega_0\zeta(2\omega_0^2(1 - 2\zeta^2) + \omega_f^2) \\ \omega_0^4(1 - 12\zeta^2 + 16\zeta^4) + \omega_0^2\omega_f^2(1 - 4\zeta^2) + \omega_f^4 \end{bmatrix} + O(\Delta t^7) \end{aligned} \quad (\text{A5})$$

where $\kappa = v_0/(\omega_0 u_0)$.